Graph Algorithms
Centrality Measures

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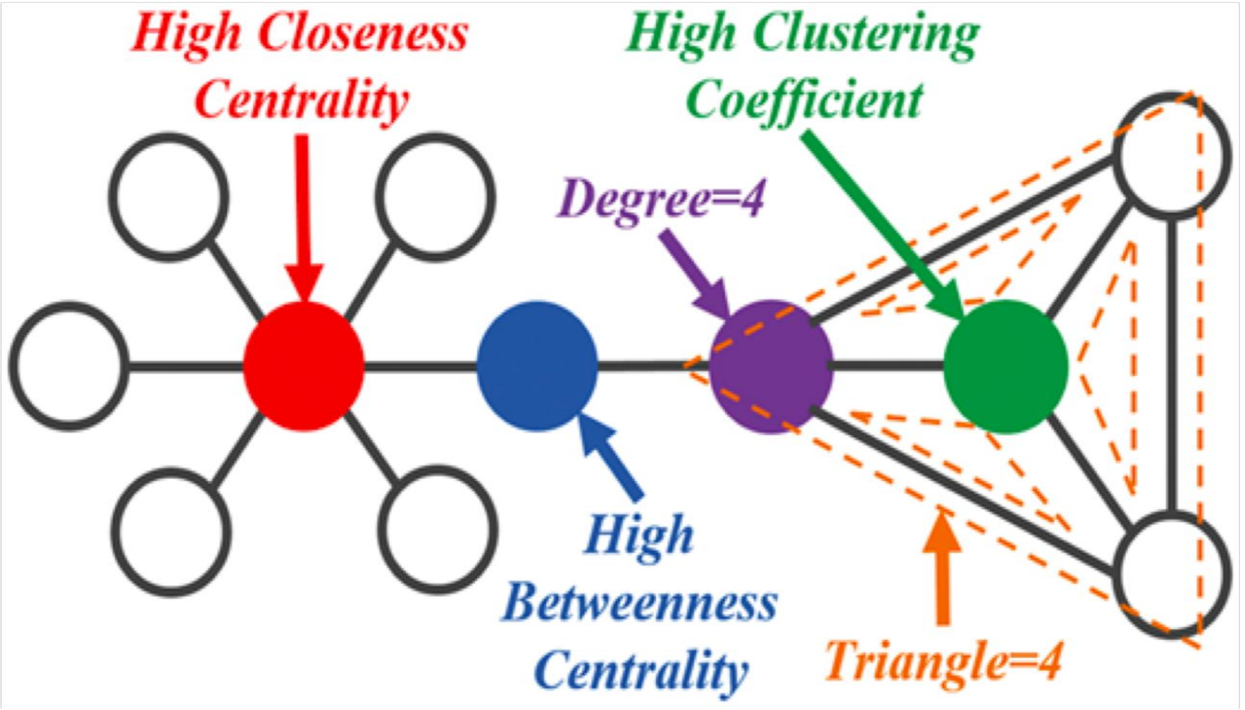
Credits

- Jure Leskovec, Stanford
- Natarajan Meghanathan, Jackson State University

Network/Graph Centrality

- Given a network, which nodes are more important or influential?
- Centrality measures reflect the importance of nodes in a network



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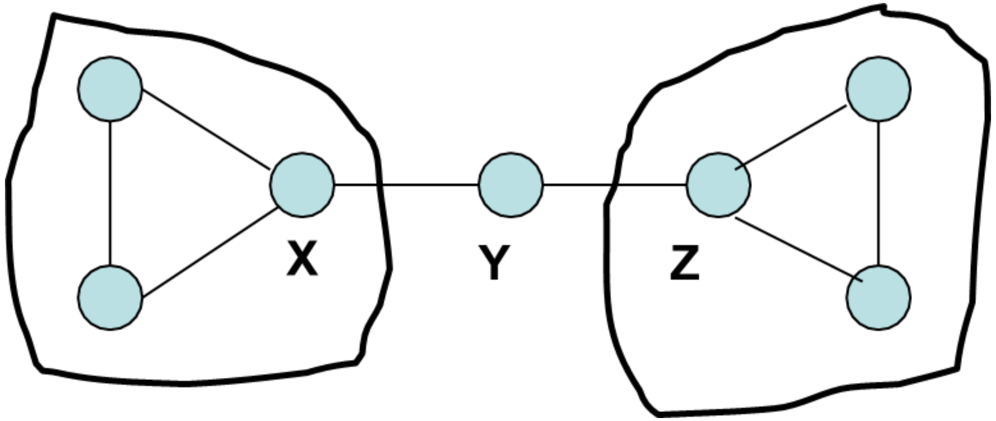
Network/Graph Centrality

- **Tells us which nodes are important in a network based on the topological structure of the network (and not information about the nodes: e.g., popularity of nodes)**
  - How influential a person is within a social network
  - Which genes play a crucial role in regulating biological systems and processes
  - Infrastructure networks: if the node is removed, how critical would it be for the functioning of the network (web/cyber, electricity grids, …)

Network/Graph Centrality

- Nodes X and Z have higher Degree
- Node Y is more central from the point of view of
  - Betweenness - to reach from one end to the other
  - Closeness – can reach every other vertex in the fewest number of hops



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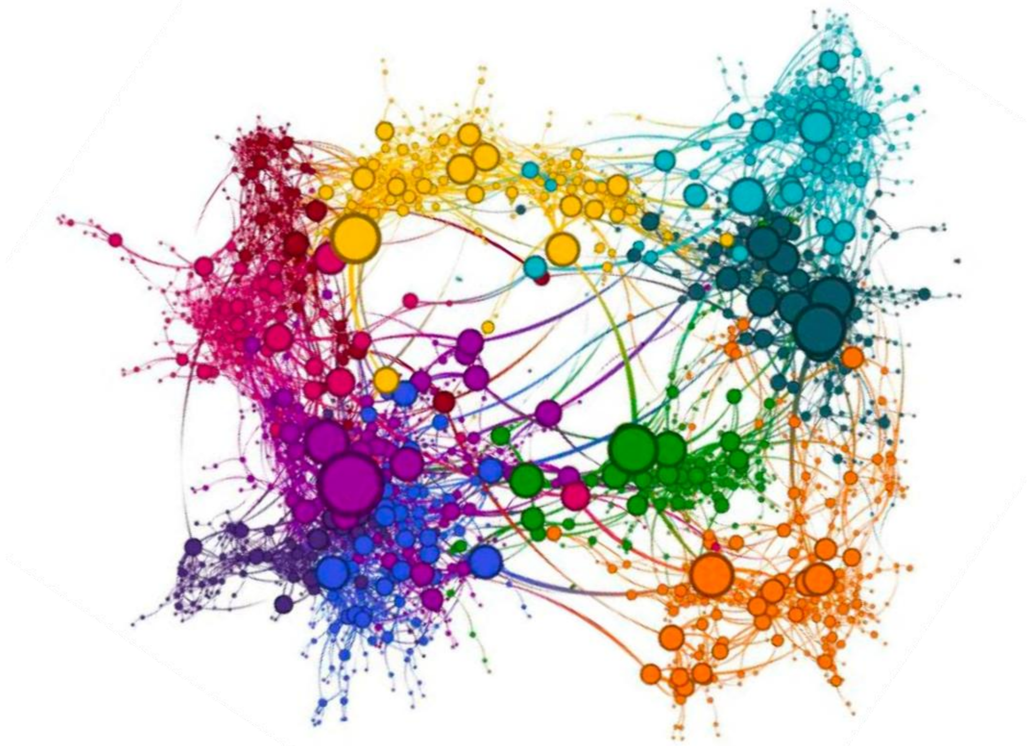
Network/Graph Centrality

- **Centrality** is used often for detecting:
  - How **influential** a person is in a social network?
  - How well used a road is in a transportation network?
  - How **important** a web page is?



[Wu and He'15]

Types of Centrality Measures

- **Geometric Measures**:
  - Importance is a function of distances to other nodes.
- **Spectral Measures**:
  - Based on the eigen-structure of some graph-related matrix
- **Path-based Measures**:
  - Take into account all (shortest) paths coming into a node

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Types of Centrality Measures

- Degree-based Centrality Metrics
  - Degree Centrality: the number of vertices adjacent to a vertex (degree)
  - Eigenvector Centrality: the degree of the vertex + the degree of its neighbors
- Shortest-path based Centrality Metrics
  - Betweenness Centrality: the number of shortest paths a node is part of
  - Closeness Centrality: measure of how close is a vertex to the other vertices [sum of the shortest path distances]
  - Farness Centrality: captures the variation of the shortest path distances of a vertex to every other vertex (eccentricity)

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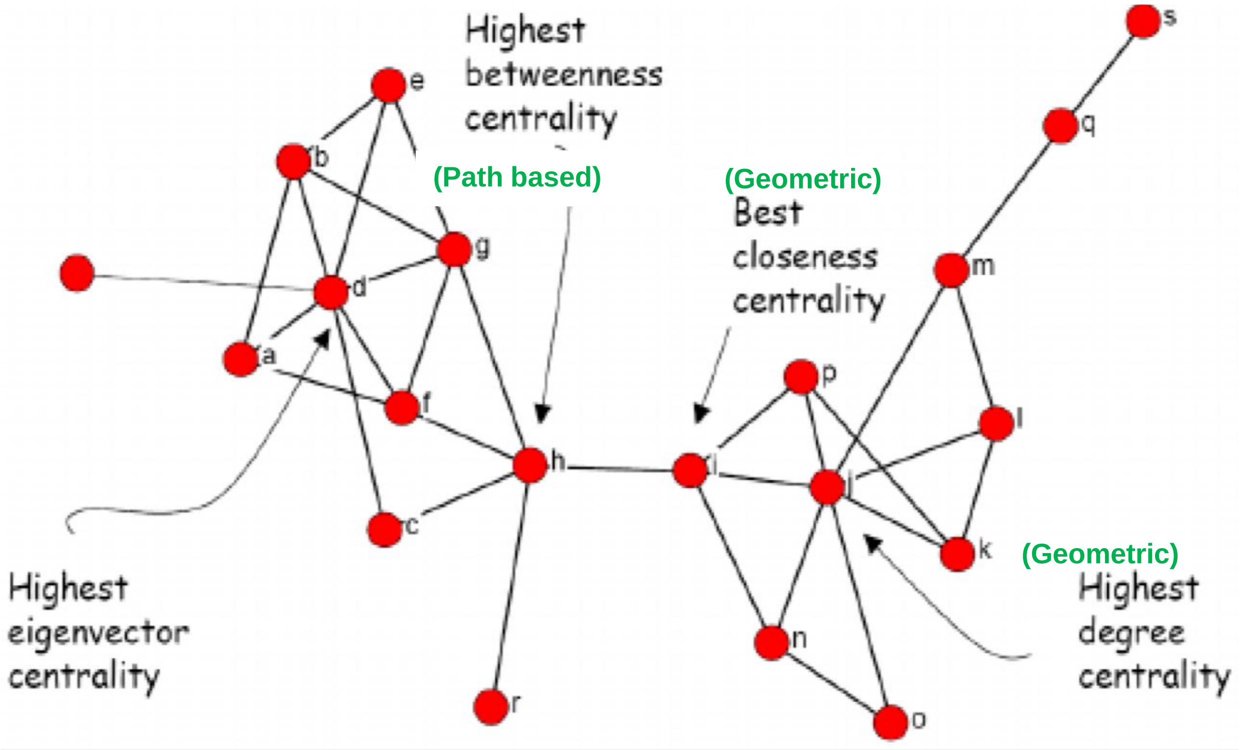
Types of Centrality Measures

- **Q: Which vertices are more central?** **A: It depends on the context**



- **Each measure identifies a different vertex as most central**
- **None is `wrong', they target different notions of importance**

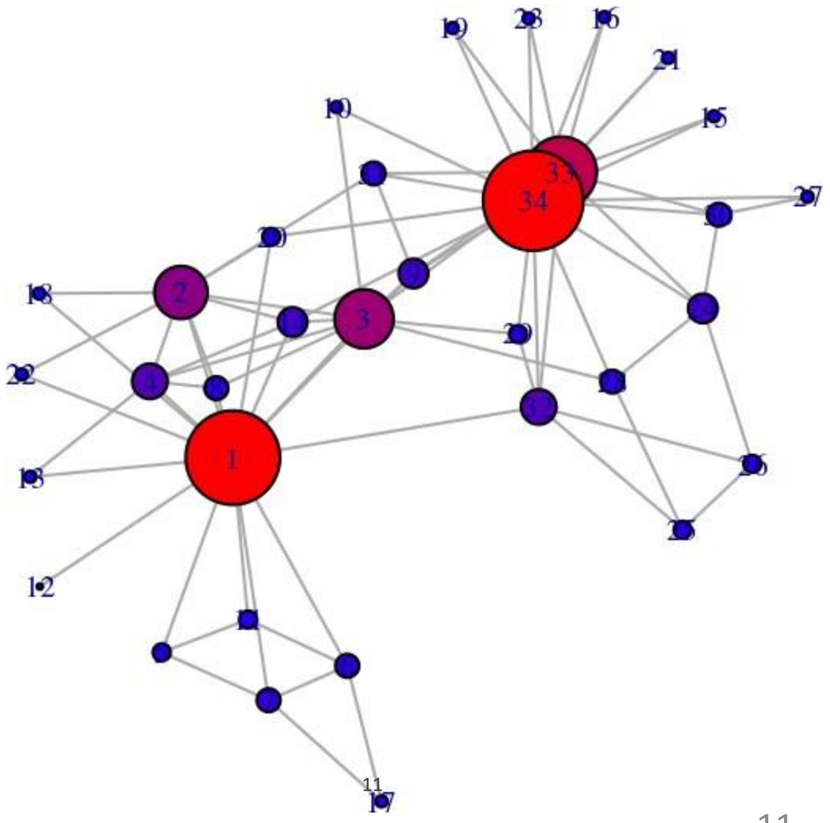
- Geometric measures of Centrality – Degree based

In-degree Centrality

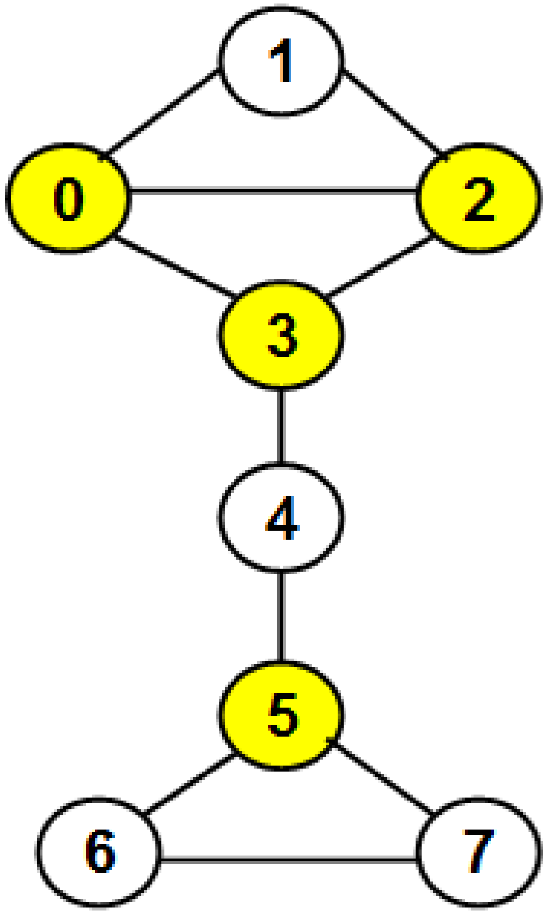
- **Geometric measures**
  - (In-)Degree Centrality: The number of incoming links

$$c_{\text{deg}}(x) = d_{in}(x)$$

  - Or equivalently, number of nodes at distance one
  - Equivalent to majority voting



Degree Centrality calculation



$$
\begin{array}{c}
\\ 0\\1\\2\\3\\4\\5\\6\\7
\end{array}
\begin{array}{c}
\begin{array}{cccccccc} 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{array} \\
\begin{bmatrix}
\boxed{0 \;\; 1 \;\; 1 \;\; 1 \;\; 0 \;\; 0 \;\; 0 \;\; 0} \\
1 \;\; 0 \;\; 1 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \\
1 \;\; 1 \;\; 0 \;\; 1 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \\
1 \;\; 0 \;\; 1 \;\; 0 \;\; 1 \;\; 0 \;\; 0 \;\; 0 \\
0 \;\; 0 \;\; 0 \;\; 1 \;\; 0 \;\; 1 \;\; 0 \;\; 0 \\
0 \;\; 0 \;\; 0 \;\; 0 \;\; 1 \;\; 0 \;\; 1 \;\; 1 \\
0 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 1 \;\; 0 \;\; 1 \\
0 \;\; 0 \;\; 0 \;\; 0 \;\; 0 \;\; 1 \;\; 1 \;\; 0
\end{bmatrix}
\end{array}
\times
\begin{bmatrix} 1\\1\\1\\1\\1\\1\\1\\1 \end{bmatrix}
=
\begin{bmatrix} \boxed{3}\\2\\3\\3\\2\\3\\2\\2 \end{bmatrix}
\begin{array}{c} \text{Vertex IDs} \\ 0\\1\\2\\3\\4\\5\\6\\7 \end{array}
$$

Adjacency Matrix — Column Vector — Degree Centrality

Time Complexity: $\Theta(V^2)$

<u>Weakness</u>: Very likely that more than one vertex has the same degree and it's not possible to uniquely rank the vertices

Degree Centrality in Directed Graphs

- **In directed graphs, we can either use the in-degree, the out degree, or the combination as the degree centrality value**

- **In practice, mostly in-degree is used.**

$$
\begin{aligned}
C_d(v_i) &= d_i^{\text{in}} \\
C_d(v_i) &= d_i^{\text{out}} \\
C_d(v_i) &= d_i^{\text{in}} + d_i^{\text{out}}
\end{aligned}
$$

d_i^{out} is the number of outgoing links for node v_i

Normalized Degree Centrality

- **Normalized by the maximum possible degree**

$$C_d^{\text{norm}}(v_i) = \frac{d_i}{n-1}$$

- **Normalized by the maximum degree**

$$C_d^{\max}(v_i) = \frac{d_i}{\max_j d_j}$$

- **Normalized by the degree sum**

$$C_d^{\text{sum}}(v_i) = \frac{d_i}{\sum_j d_j} = \frac{d_i}{2|E|}$$

- **Works for directed/undirected graphs**

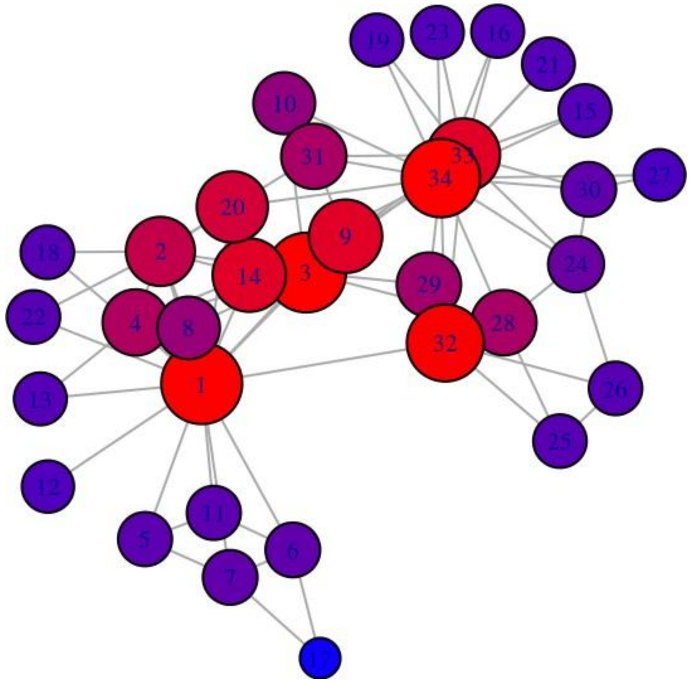
Why bother normalizing?

Closeness Centrality

- **Geometric measures**
  - Closeness Centrality: Who are the bridges?
  - Nodes that are more central have smaller distances
  - Nodes that are more central have smaller distances, and higher centrality

$$c_{\mathrm{clos}}(x) = \frac{1}{\sum_y \boxed{d(y,x)}}$$

$d(y,x)$ length of the shortest path from y to x

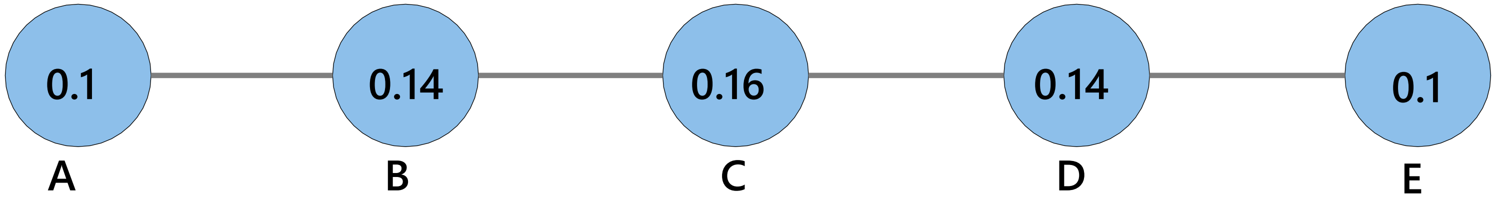


Closeness Centrality

- **Geometric measures**

- How much a vertex can communicate without relying on third parties for his messages to be delivered

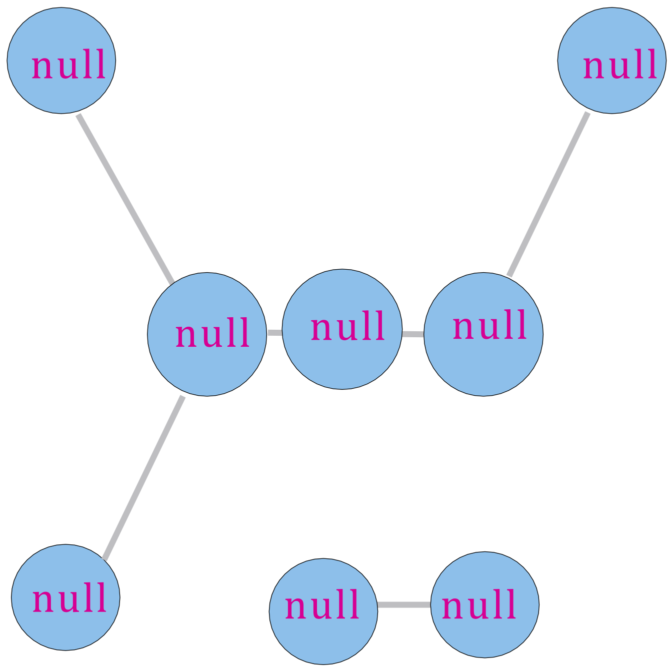
C is the most central

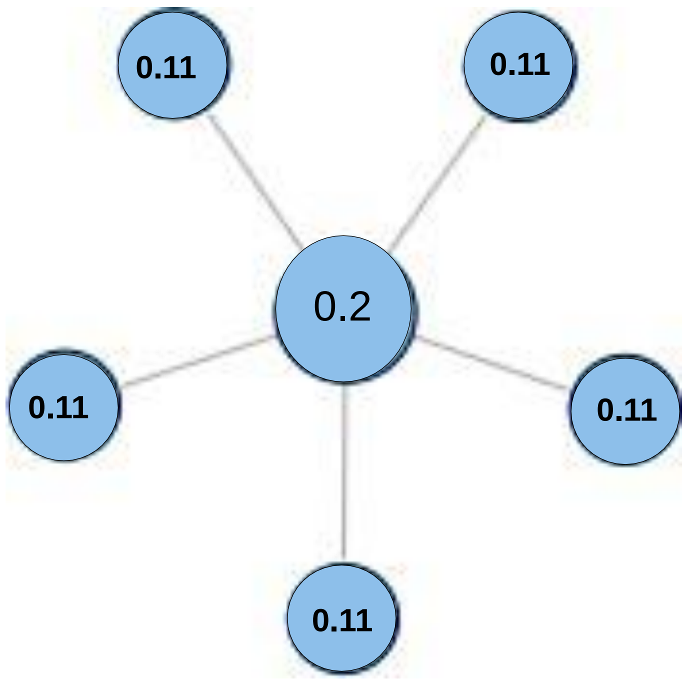


$$c_{clos}(A) = \frac{1}{1+2+3+4} = 0.1$$

Problem: The graph must be **(strongly) connected!**

Example





We get null score for all nodes, if the graph is not connected!

Harmonic Centrality

- **Harmonic Centrality**: Who are the **bridges**?
  - Replace the average distance with the harmonic mean of all distances.
  - The $n(n-1)$ distances between every pair of distinct nodes:

solve the problem of dealing with unconnected graphs

$$c_{\mathrm{har}}(x) = \boxed{\sum_{y \neq x} \frac{1}{d(y,x)}} = \sum_{d(y,x)<\infty, y \neq x} \frac{1}{d(y,x)}$$

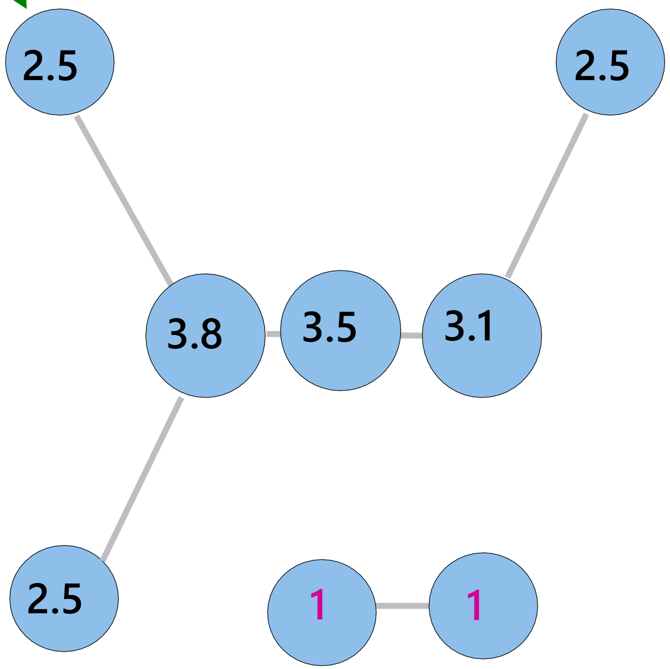
Harmonic mean

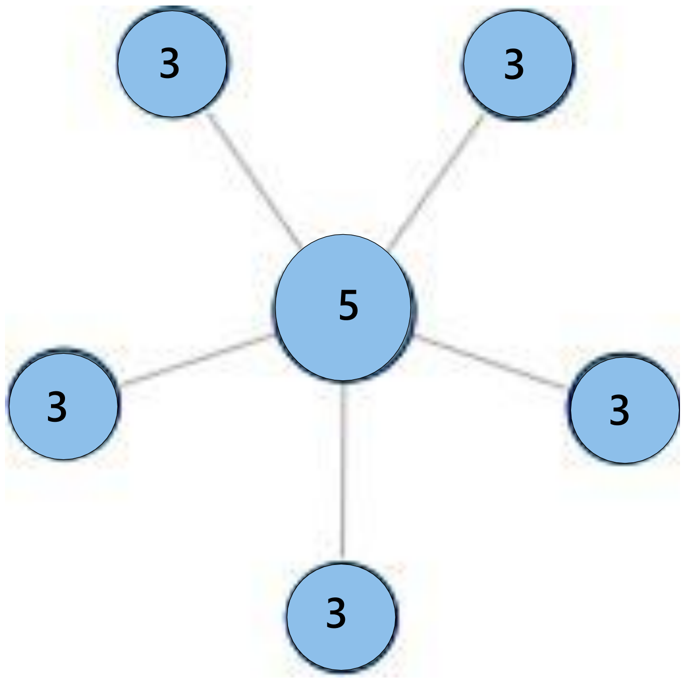
$$c_{\mathrm{clos}}(x) = \frac{1}{\sum_y d(y,x)}$$

  - Strongly correlated to closeness centrality
  - Naturally also accounts for nodes y that cannot reach x
  - Can be applied to graphs that are **not strongly connected**

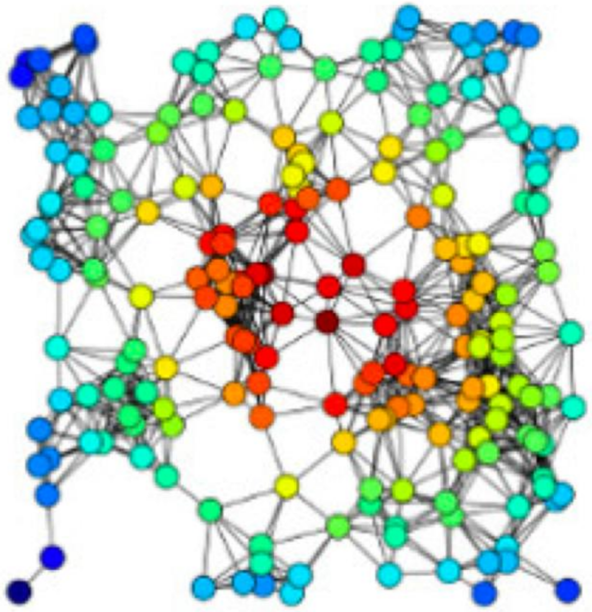
Example

$$c_{harm} = \frac{1}{1} + \frac{1}{2} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} = 2.58$$

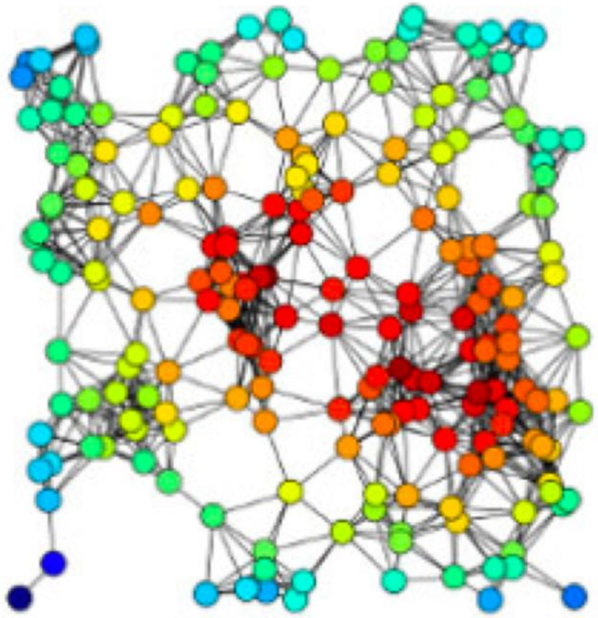




Closeness vs. Harmonic Centrality



Red nodes are closer to all the other nodes



Red nodes are closer to all the other nodes, and have larger degrees

Closeness centrality and Harmonic Centrality of the same graph. 20

- **Spectral Measures of Centrality**

Eigenvalues and eigenvectors Intro

- **Eigenvalue – (ע"ע) ערך עצמי   Eigenvector – (ו"ע) וקטור עצמי**

The Eigenvalue problem

Let $\boldsymbol{A}$ be an $n \times n$ matrix:

$\boldsymbol{x} \neq \boldsymbol{0}$ is an **eigenvector** of $\boldsymbol{A}$ if there exists a scalar λ such that

$$\boldsymbol{A}\,\boldsymbol{x} = \lambda\,\boldsymbol{x}$$

where λ is called an **eigenvalue**.

Eigenvalues and eigenvectors Intro

- **How do we find eigenvalues?**
- **The linear algebra approach**

$$\boldsymbol{A}\,\boldsymbol{x} = \lambda\,\boldsymbol{x}$$
$$(\boldsymbol{A} - \lambda\,\boldsymbol{I})\boldsymbol{x} = \boldsymbol{0}$$

Therefore the matrix $(\boldsymbol{A} - \lambda\,\boldsymbol{I})$ is singular $\Longrightarrow det(\boldsymbol{A} - \lambda\,\boldsymbol{I}) = 0$

$p(\lambda) = det(\boldsymbol{A} - \lambda\,\boldsymbol{I})$ is the characteristic polynomial of degree n.

Eigenvalues and eigenvectors Intro

- **Example**

$$\boldsymbol{A - \lambda I}$$

$$\boldsymbol{A} = \begin{pmatrix} 2 & 1 \\ 4 & 2 \end{pmatrix} \qquad det \begin{pmatrix} 2-\lambda & 1 \\ 4 & 2-\lambda \end{pmatrix} = 0 \qquad \boldsymbol{(2-\lambda)(2-\lambda) - 1 \cdot 4 = 0}$$

Solution of characteristic polynomial gives: $\lambda_1 = 4, \lambda_2 = 0$

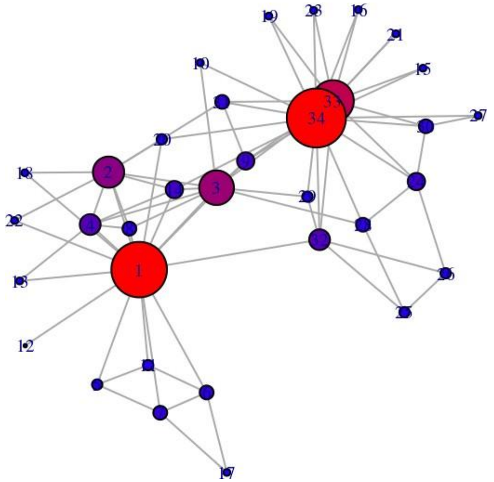
To get the eigenvectors, we solve: $\boldsymbol{A}\,\boldsymbol{x} = \lambda\,\boldsymbol{x}$

$$\lambda_1 = 4 \qquad \begin{pmatrix} 2-(4) & 1 \\ 4 & 2-(4) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad \boldsymbol{x} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

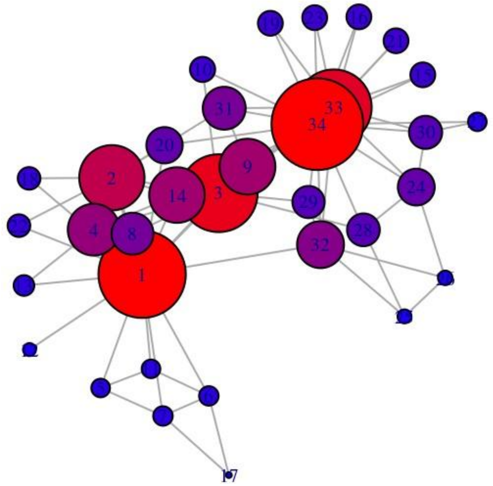
$$\lambda_1 = 0 \qquad \begin{pmatrix} 2-(0) & 1 \\ 4 & 2-(0) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \qquad \boldsymbol{x} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$$

Spectral Centrality

- Spectral measures
  - Compute the dominant eigenvector of a matrix derived from the graph
  - Idea: A node's centrality is a function of the centrality of its neighbors
    - Nodes connected to central nodes has a larger centrality score than those connected to non-central nodes (tell me who your friends are)
- Types:
  - Eigenvector Centrality
  - Page Rank
  - Hits



Degree centrality



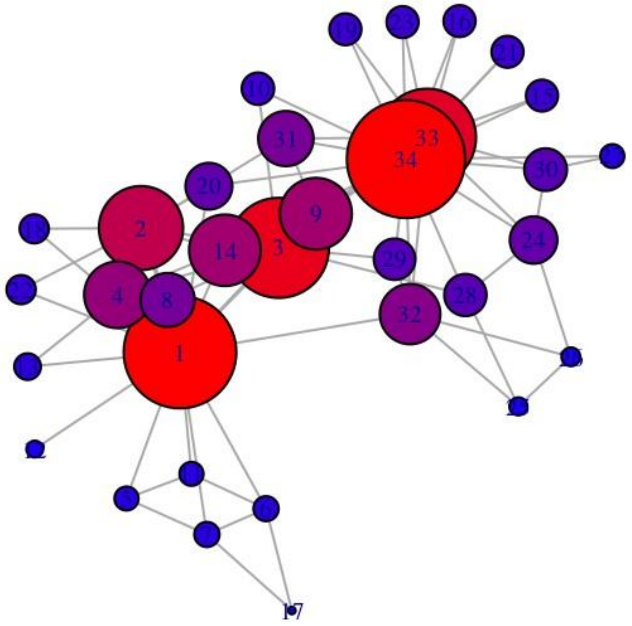
Eigenvector centrality

Eigen (vector) -centrality

- **Spectral measures**
  - **Eigenvector Centrality**: Measure of the **influence** of a node in a network
  - **Idea:** Every node starts with the same score, and then each node gives away its score to its successors

$$c_{\text{eig}}(x) = \frac{1}{\lambda} \sum_{y \to x} c_{\text{eig}}(y)$$

Normalization constant = $\|c_{eig}\|_2$



- **Intuitively**:
  Degree centrality counts walks of length one;
  Eigenvalue centrality counts walks of length infinity

Eigencentrality

- **Eigenvector Centrality** **measure of the influence of a node in a network:**

$$c_{\mathrm{eig}}(x) = \frac{1}{\lambda} \sum_{y \to x} c_{\mathrm{eig}}(y)$$

  - c_{eig} converges to the dominant eigenvector of adj. matrix A
  - λ converges to the dominant eigenvalue of adj. matrix A

- **Equivalently, eigencentrality is the eigenvector correspond to the dominant eigenvalue** (λ) of A **(adj. matrix)**

$$AX = \lambda X$$

- **Problem**: Graph should be **strongly connected**!

Eigencentrality calculation

- **Power Iteration:**

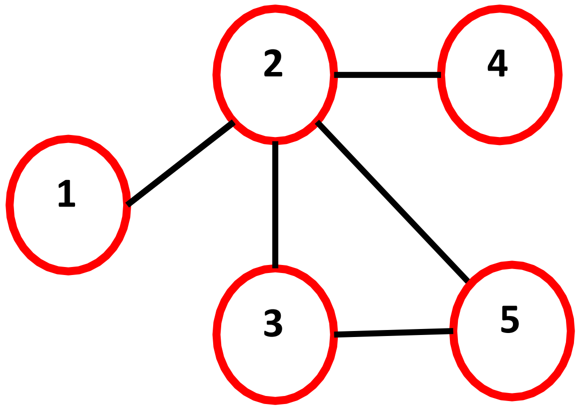
Set $\boldsymbol{c}^{(0)} \leftarrow 1, k \leftarrow 1$

1: $\boldsymbol{c}^{(k)} \leftarrow A\boldsymbol{c}^{(k-1)}$

2: $\boldsymbol{c}^{(k)} = \boldsymbol{c}^{(k)} / \|\boldsymbol{c}^{(k)}\|_2$

3: If $\|\boldsymbol{c}^{(k)} - \boldsymbol{c}^{(k-1)}\| > \varepsilon$:

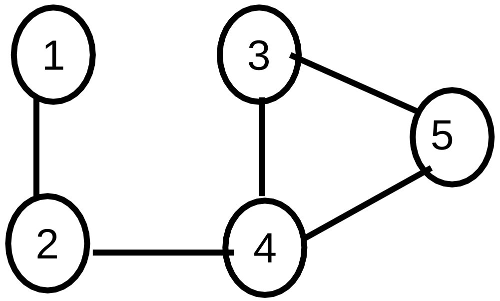
4: $k \leftarrow k + 1$, goto **1**



$$A = \begin{array}{c} \begin{array}{ccccc} 1 & 2 & 3 & 4 & 5 \end{array} \\ \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix} \end{array} \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{array} \qquad c = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$$

Time Complexity: $\Theta(V^3)$

EigenVector Centrality Example



$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

Let $C_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

Iteration 1

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 2 \\ 3 \\ 2 \end{bmatrix} \equiv \begin{bmatrix} 0.213 \\ 0.426 \\ 0.426 \\ 0.639 \\ 0.426 \end{bmatrix}$$

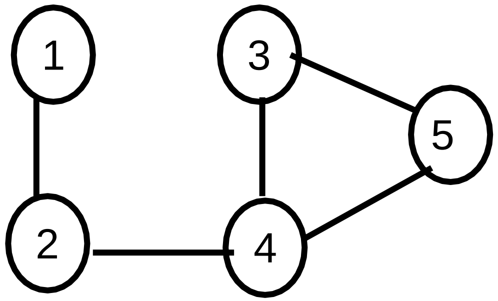
Normalized Value = 4.69

Iteration 2

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0.213 \\ 0.426 \\ 0.426 \\ 0.639 \\ 0.426 \end{bmatrix} = \begin{bmatrix} 0.426 \\ 0.852 \\ 1.065 \\ 1.278 \\ 1.065 \end{bmatrix} \equiv \begin{bmatrix} 0.195 \\ 0.389 \\ 0.486 \\ 0.584 \\ 0.486 \end{bmatrix}$$

Normalized Value = 2.19

EigenVector Centrality Example



$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

Let X0 = $\begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$

<u>Iteration 3</u>

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0.195 \\ 0.389 \\ 0.486 \\ 0.584 \\ 0.486 \end{bmatrix} = \begin{bmatrix} 0.389 \\ 0.779 \\ 1.07 \\ 1.361 \\ 1.07 \end{bmatrix} \equiv \begin{bmatrix} 0.176 \\ 0.352 \\ 0.484 \\ 0.616 \\ 0.484 \end{bmatrix}$$

Normalized Value = 2.21

<u>Iteration 4</u>

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0.176 \\ 0.352 \\ 0.484 \\ 0.616 \\ 0.484 \end{bmatrix} = \begin{bmatrix} 0.352 \\ 0.792 \\ 1.100 \\ 1.320 \\ 1.100 \end{bmatrix}$$

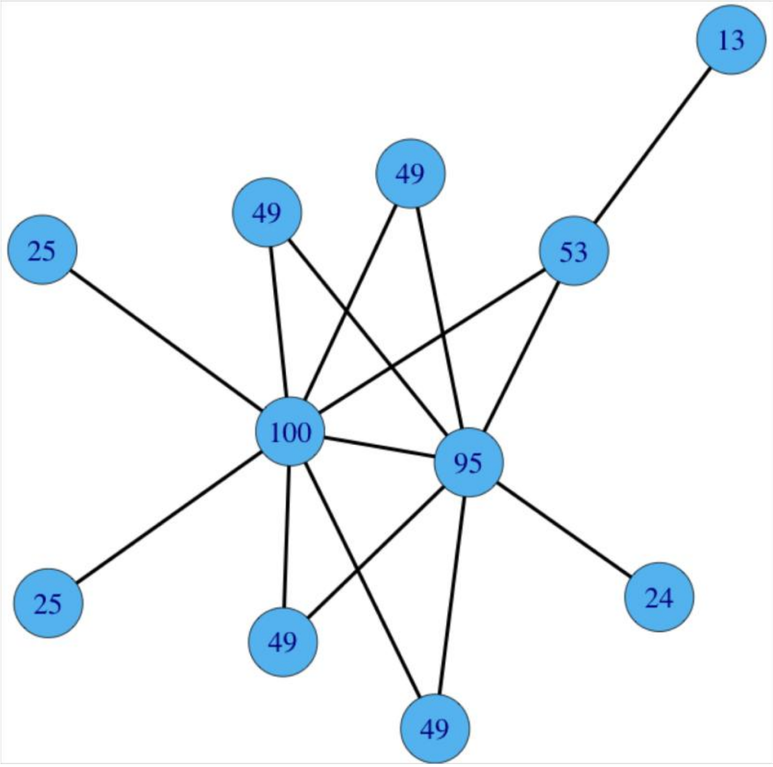
Normalized Value = 2.21 converges

<u>Eigen Vector Centrality</u>

$$\begin{array}{cc} 1 & \\ 2 & \\ 3 & \\ 4 & \\ 5 & \end{array} \begin{bmatrix} 0.176 \\ 0.352 \\ 0.484 \\ 0.616 \\ 0.484 \end{bmatrix}$$

Converged

Example



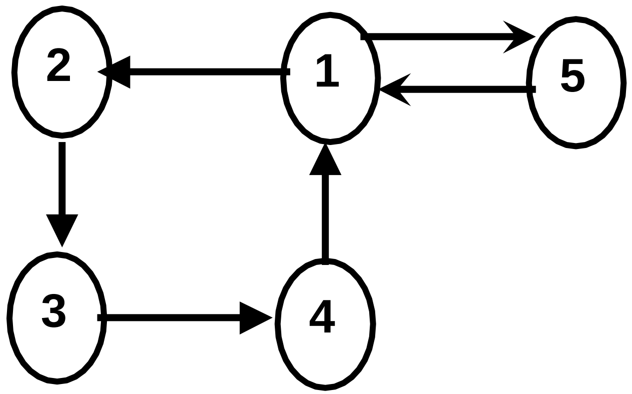
Eigenvalue centrality counts walks of length infinity

Each matrix multiplication adds contribution from more distant nodes

Eigen Vector Centrality for Directed Graphs

- For directed graphs, we can use the Eigenvector centrality to evaluate:
  - "importance" of a node (based on the out-degree Eigen Vector)
  - "prestige" of a node (based on the in-degree Eigen Vector)
- A node is considered to be more important if it has out-going links to nodes that in turn have a larger out-degree (i.e., more out-going links).
- A node is considered of higher "prestige", if it has in-coming links from nodes that themselves have a larger in-degree (i.e., more in-coming links).

Eigen Vector Centrality for Directed Graphs



$$\begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Out-going links based Adj. Matrix

Importance of Nodes (Out-deg. Centrality)

Node	Score
1	0.5919
4	0.4653
5	0.4653
3	0.3658
2	0.2876

$$\begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

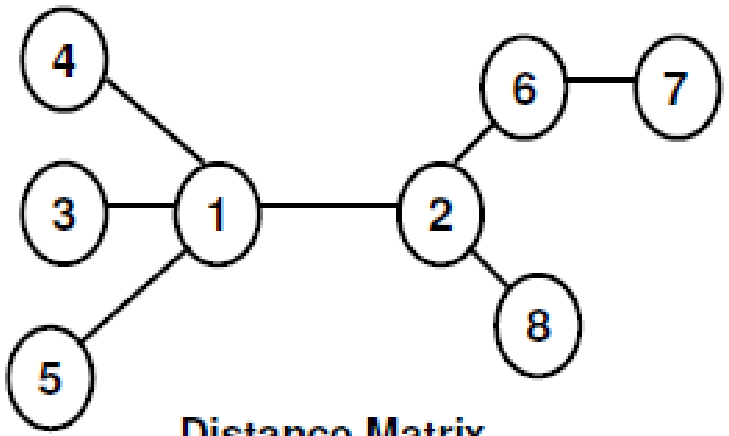
In-coming links based Adj. Matrix

Prestige of Nodes (In-deg. Centrality)

Node	Score
1	0.5919
2	0.4653
5	0.4653
3	0.3658
4	0.2876

Different Nodes order, same score

Closeness and Farness Centrality



Distance Matrix

	1	2	3	4	5	6	7	8	Sum of distances	Closeness
1	0	1	1	1	1	2	3	2	11	0.0909
2	1	0	2	2	2	1	2	1	11	0.0909
3	1	2	0	2	2	3	4	3	17	0.0588
4	1	2	2	0	2	3	4	3	17	0.0588
5	1	2	2	2	0	3	4	3	17	0.0588
6	2	1	3	3	3	0	1	2	15	0.0667
7	3	2	4	4	4	1	0	3	21	0.0476
8	2	1	3	3	3	2	3	0	17	0.0588

Time Complexity: $\Theta(VE + V^2)$

$$C_{farn}(x) = \sum_y d(y,x)$$

$$C_{clos}(x) = 1/C_{farn}(x)$$

$$C_{clos}(x) = \frac{1}{\sum_y \boxed{d(y,x)}}$$

Katz's index/measure (Spectral)

- **Katz's Index/measre: Measures influence by taking into account the total number of walks between a pair of nodes**

$$c_{\text{katz}}(x) = \boldsymbol{\beta} + \sum_{k=0}^{\infty} \sum_{x \to y} \alpha^k \boxed{\left(A^k\right)_{xy}}$$

$$x = \alpha A x + \beta$$

Total number of walks of length k between x, y

  - α is an attenuation factor in range $\left(0, \frac{1}{\lambda}\right)$, where λ is the largest eigenvalue of A
  - β is to give some nodes more privilege
  - **Long paths are weighted less than short ones**

How to weight the contribution of 'further away' nodes (long paths)

Katz's Index example

$$c_{\text{katz}}(x) = \boldsymbol{\beta} + \sum_{k=0}^{\infty} \sum_{x \to y} \boxed{\alpha^k} \boxed{(A^k)_{xy}}$$

$\alpha < 1$: Long paths are weighted less

Total number of walks of length k between x, y

$A^0 = \mathbf{I}$

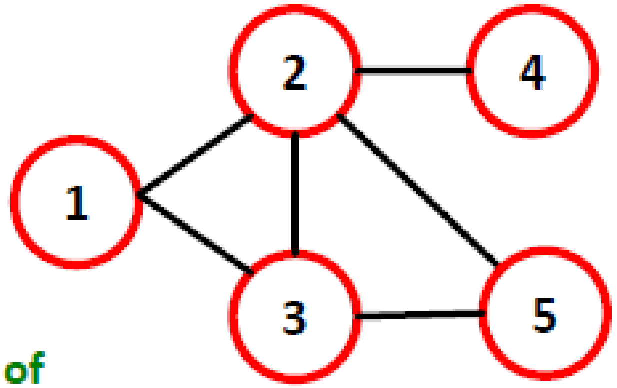
$$A^1 = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 2 & 1 & 1 & 1 & 2 \\ 1 & 4 & 2 & 0 & 1 \\ 1 & 2 & 3 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 \\ 2 & 1 & 1 & 1 & 2 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}^3 = \begin{bmatrix} 2 & 6 & 5 & 1 & 2 \\ 6 & 4 & 6 & 4 & \boxed{6} \\ 5 & 6 & 4 & 2 & 5 \\ 1 & 4 & 2 & 0 & 1 \\ 2 & 6 & 5 & 1 & 2 \end{bmatrix}$$

Number of walks of length 3 between 2, 5

(2,1,3,5), (2,4,2,5), (2,3,2,5),
(2,1,2,5), (2,5,3,5), (2,5,2,5)



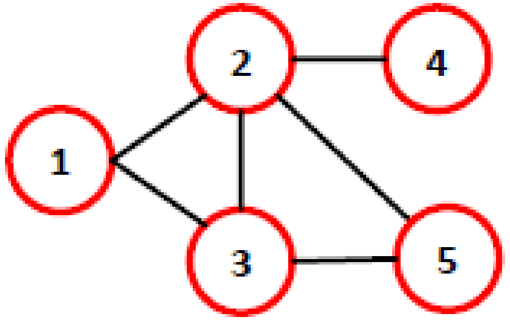
Katz's Index Calculation

- **Katz's Index**: Give each node a small amount of centrality β for free

$$c_{\text{katz}}(x) = \alpha \sum_{y \to x} (c_{\text{Katz}}(y) + \beta)$$

Normalization constant

- **Power Iteration:**
  - Set $\boldsymbol{c}^{(0)} \leftarrow 1, k \leftarrow 1$
  - **1:** $\boldsymbol{c}^{(k)} \leftarrow \alpha \boldsymbol{A} \boldsymbol{c}^{(k-1)} + \beta \mathbf{1}$
  - **2:** If $\left\| \boldsymbol{c}^{(k)} - \boldsymbol{c}^{(k-1)} \right\| > \varepsilon$:
  - **3:** $\quad k \leftarrow k + 1$, goto **1**



Katz's Index example

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix}$$

- Small constant: $\alpha = 0.1$
- Initial centrality vector: $x^{(0)} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$
- Bias vector: $\beta = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$

Calculate $x^{(1)} = 0.1\mathbf{A}x^{(0)} + \beta$

$$\mathbf{A}x^{(0)} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 1 \\ 2 \\ 2 \end{pmatrix}$$

$$0.1\mathbf{A}x^{(0)} = \begin{pmatrix} 0.2 \\ 0.3 \\ 0.1 \\ 0.2 \\ 0.2 \end{pmatrix}$$

$$x^{(1)} = \begin{pmatrix} 0.2 \\ 0.3 \\ 0.1 \\ 0.2 \\ 0.2 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1.2 \\ 1.3 \\ 1.1 \\ 1.2 \\ 1.2 \end{pmatrix}$$

Katz's Index example

2'nd iteration **Calculate** $x^{(2)} = 0.1\mathbf{A}x^{(1)} + \beta$

$$\mathbf{A}x^{(1)} = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1.2 \\ 1.3 \\ 1.1 \\ 1.2 \\ 1.2 \end{pmatrix} = \begin{pmatrix} 2.4 \\ 3.7 \\ 1.2 \\ 2.5 \\ 2.5 \end{pmatrix}$$

$$0.1\mathbf{A}x^{(1)} = \begin{pmatrix} 0.24 \\ 0.37 \\ 0.12 \\ 0.25 \\ 0.25 \end{pmatrix}$$

$$x^{(2)} = \begin{pmatrix} 0.24 \\ 0.37 \\ 0.12 \\ 0.25 \\ 0.25 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1.24 \\ 1.37 \\ 1.12 \\ 1.25 \\ 1.25 \end{pmatrix}$$

$$\Longrightarrow \quad x^{(\infty)} \approx \begin{pmatrix} 1.2503 \\ 1.3779 \\ 1.1250 \\ 1.2642 \\ 1.2642 \end{pmatrix}$$

Node 2 is the most influential (1.378)

Node 3 is the least influential (1.125)

Katz's Index properties

- Katz's Index: Suitable for directed acyclic graphs
- How to choose a?
  - For a close to 0, the contribution given by paths longer than one rapidly declines, and thus Katz scores are mainly influenced by short paths (mostly in-degrees)
  - When the a is large, long paths are devalued smoothly, and Katz scores are more influenced by the topology of the network
  - Katz measure diverges at $\alpha > 1/\lambda$ (the dominant eigenvalue)
  - The dominant eigenvector of A is the limit of Katz centrality as α approaches $1/\lambda$

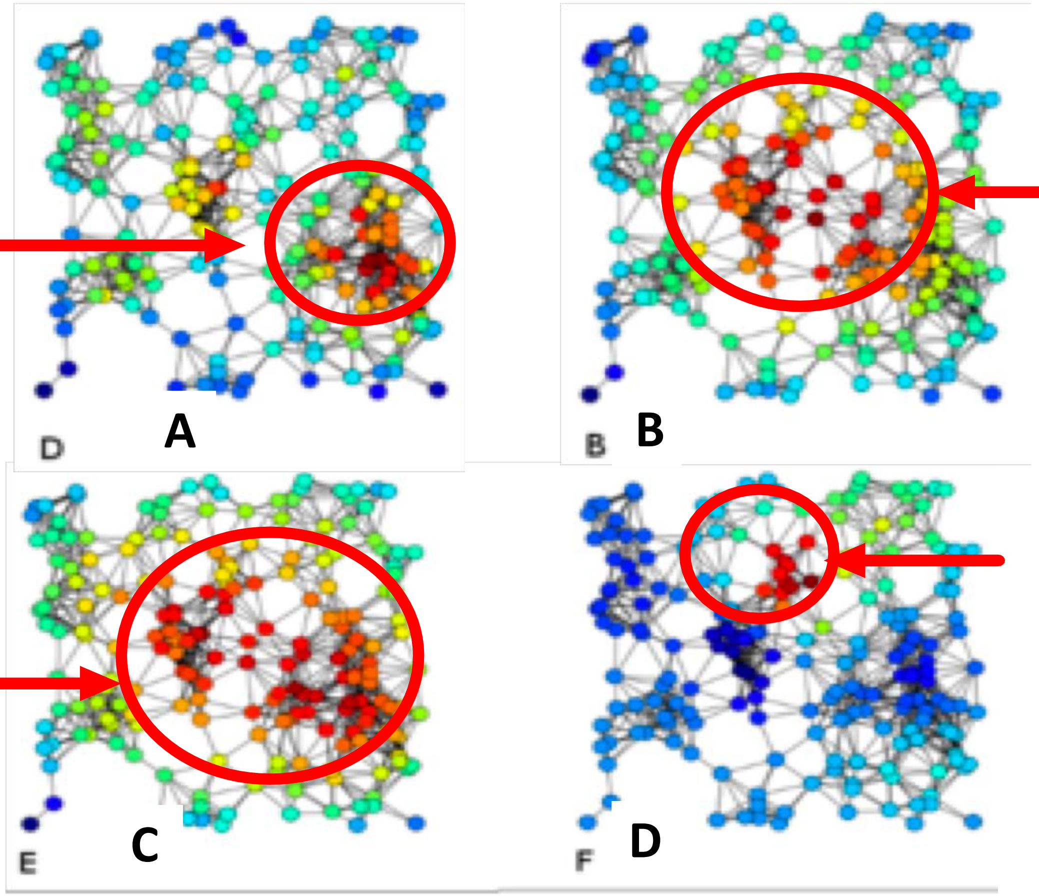
Large networks centrality examples

They have largest degrees

They are closer to all the other nodes

They are closer to all the other nodes, and have larger degrees

Lots of paths passing through these nodes

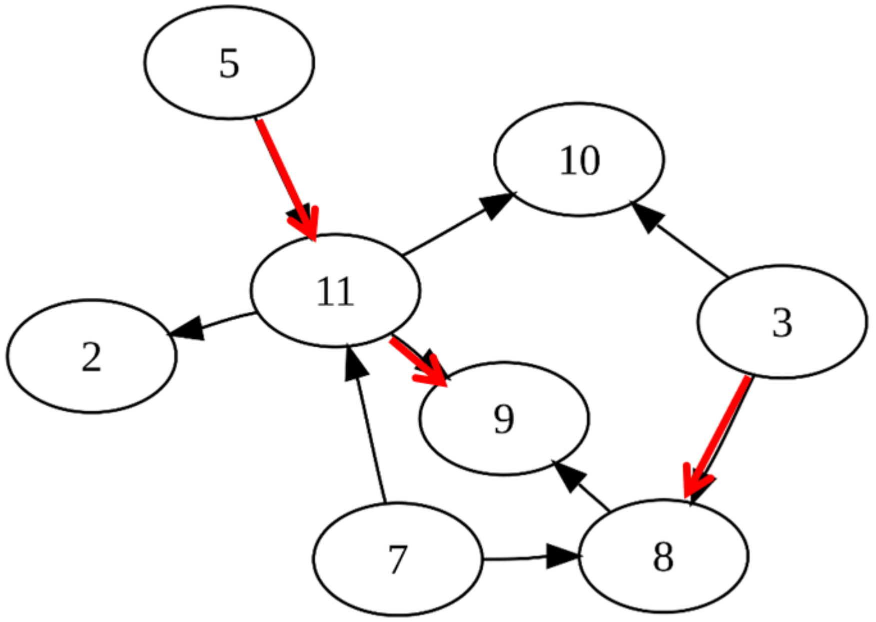


A) Degree, B) Closeness, C) Harmonic, D) Katz centrality of the same graph.

Page Rank Algorithm

Page Rank

- In directed graphs: some in-degrees are zero.
- Fix the issue: Katz centrality used a "free" weight of β
- <u>New problem</u>: should the weight of the following edges be the same:
  (11, 9),
  (5, 11),
  (3, 8)?



- How should we decide on the weight?

Web search methods

- Early search engines mainly compared content similarity of the query and the indexed pages. i.e.,
  - They use information retrieval methods, cosine similarity, …
- In the mid 1990's, it became clear that content similarity alone was no longer sufficient.
  - The number of pages grew rapidly in the mid 1990's.
    - How to choose only 30-40 pages and rank them suitably to present to the user?
  - Content similarity is easily spammed
    - Webpage can repeat words and add related words to boost the rankings of his pages and/or to make the pages relevant to a large number of queries.

Web search methods

- Starting around 1996, researchers began to work on the problem. They resorted to hyperlinks.
- Web pages are connected through hyperlinks, which carry important information.
  - Hyperlinks may organize information at the same site (anchors).
  - Hyperlinks can point to pages from other Web sites. Such out-going hyperlinks often convey authority to the pages being pointed to.
- Those pages that are pointed to by many other pages are likely to contain authoritative information.

Web search methods

- Around 1997-1998, two most influential hyperlink based search algorithms PageRank and HITS were published.
- Both algorithms exploit the hyperlinks of the Web to rank pages according to their levels of “prestige” or “authority”.
  - HITS : Jon Kleinberg (Hypertext Induced Topic Search)
  - PageRank : Sergey Brin and Larry Page, PhD students
- PageRank has emerged as the dominant link analysis model
  - Due to its query-independence, ability to combat spamming, and
  - Google’s huge business success.

PageRank: intuitive idea

- PageRank relies on the democratic nature of the Web by using its huge link structure as an indicator of an individual page's value
- PageRank interprets a hyperlink from page *i* to page *j* as a vote, by page *i*
- However, PageRank looks at more than the number of votes; it also analyzes the page that casts the vote.

1. A vote by an "important" page *i* weighs more and helps to make page *j* more "important." (like eigenvector and Katz)
2. Also, the vote of page *i* is shared among the pages that it points to, so page *j* gets a fraction of the vote.

PageRank: intuitive idea

- A hyperlink from a page to another page is an implicit transmission of authority to the target page
  - The more in-links that a page *i* receives, the more prestige the page *i* has.
- Pages that point to page *i* also have their own prestige scores
  - A page of a higher prestige pointing to *i* is more important than a page of a lower prestige pointing to *i*.
  - In other words, **a page is important if it is pointed to by other important pages.**

Computing PageRank

- **PageRank (PR) original formula:**

$$PR(u) = \sum_{v \in B_u} \frac{PR(v)}{L(v)}$$

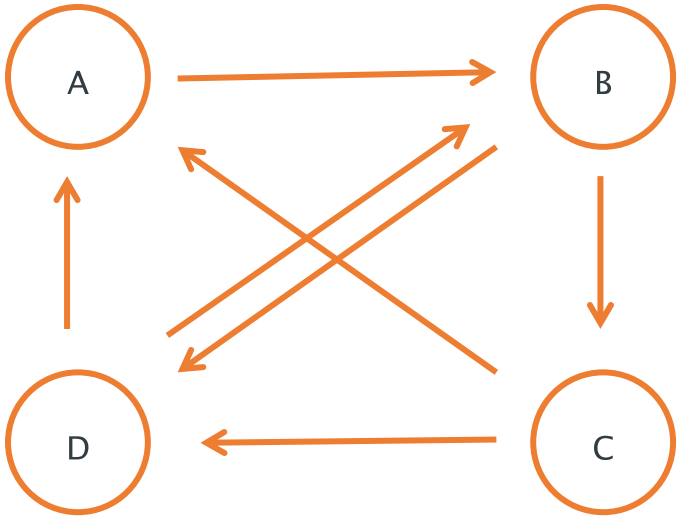
(B_u - the set containing all pages linking to page **u**)

$L(v)$ – number of links outgoing from **V**

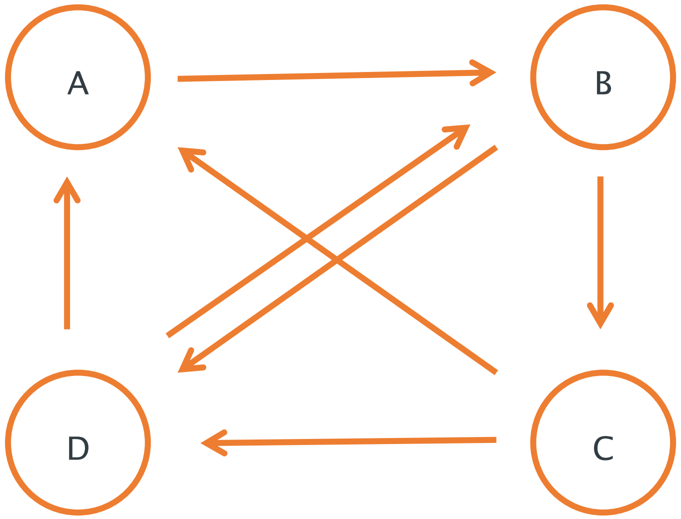
- **PageRank (*PR*) of page u is given by the summation of the PR of all pages in the set of pages linking to page *u* ($v \in B_u$), divided by the number *L(v)* of links from page *v* (outgoing degree)**
- **Iterative formula, starting with rank *1/n* for all *n* pages (nodes)**

Simple example

- **Simple example of 4 linked pages (nodes)**



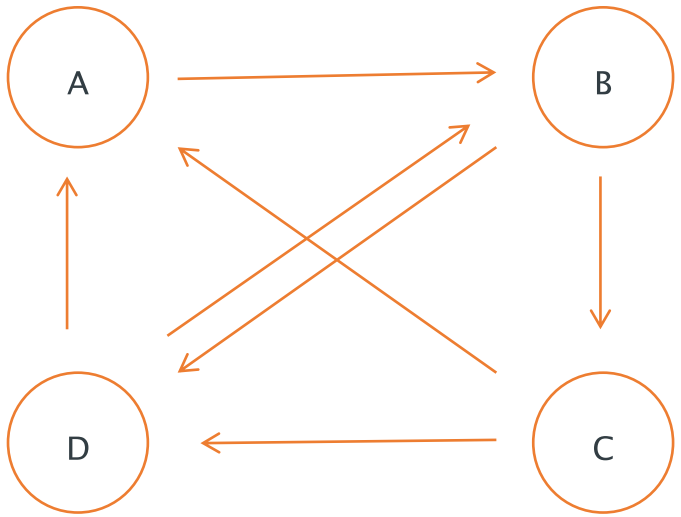
Pages and Links



	Iteration 1
A	1/n = 1/4
B	1/n = 1/4
C	1/n = 1/4
D	1/n = 1/4

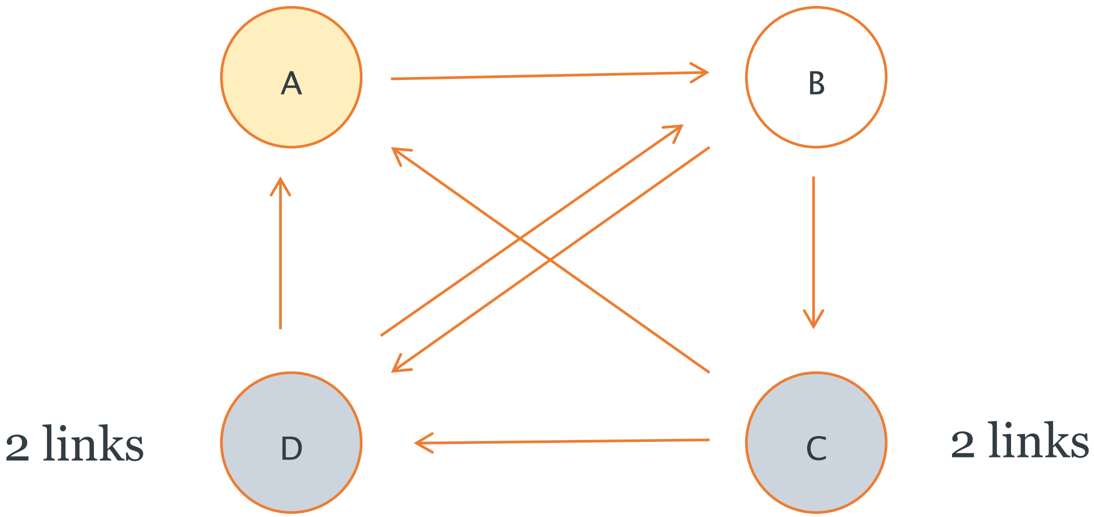
First iteration ranks all pages *1/n*, where *n* is the number of pages

Cont.



	Iteration 1
A	1/4
B	1/4
C	1/4
D	1/4

Cont.

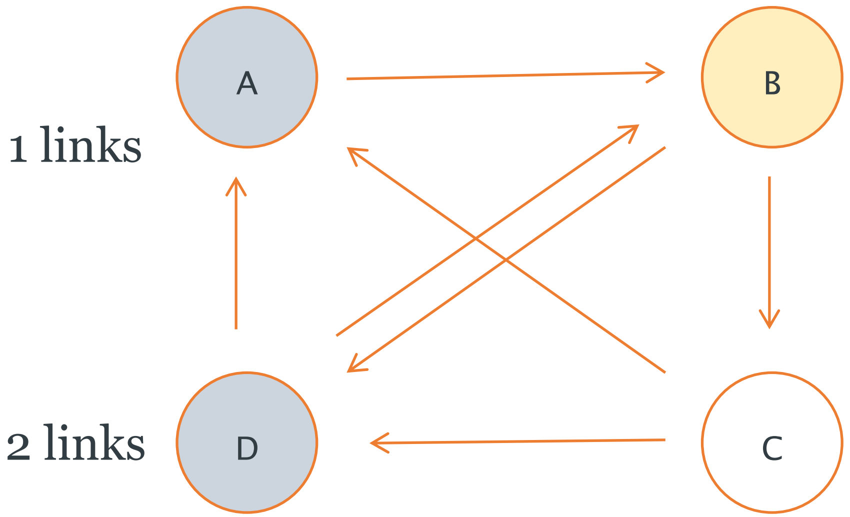


$$PR(u) = \sum_{v \in B_u} \frac{PR(v)}{L(v)}$$

	Iteration 1	Iteration 2
A	1/4	¼ / 2 + ¼ / 2 = 2/8
B	1/4	
C	1/4	
D	1/4	

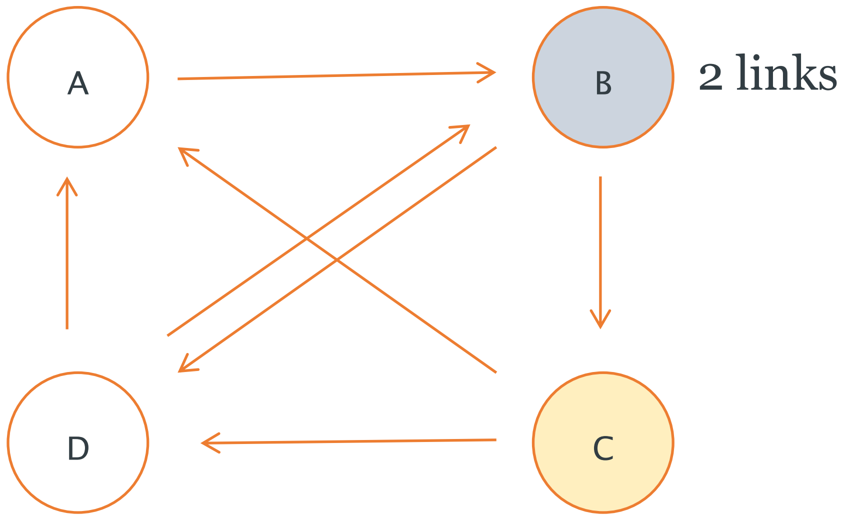
$$PR^{2}(u) = \frac{\sum PR^{1}(v)}{L(v)}$$

Cont.



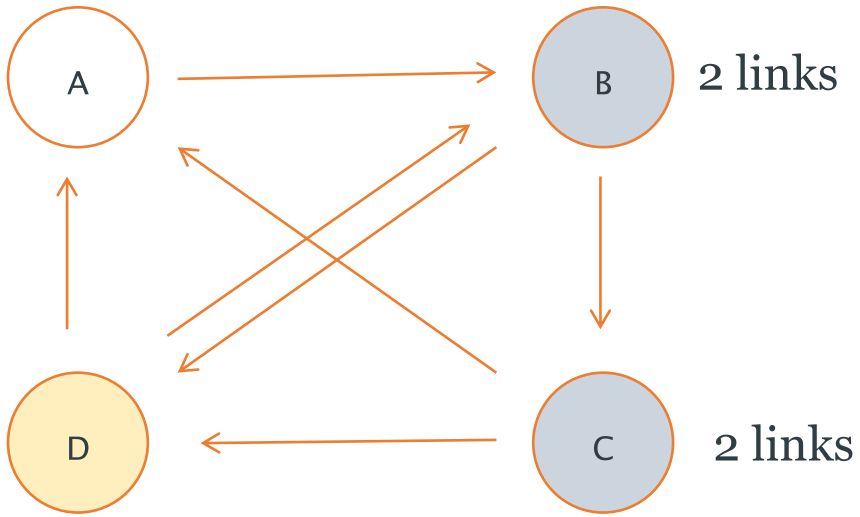
	Iteration 1	Iteration 2
A	1/4	¼ / 2 + ¼ / 2 = 2/8
B	1/4	¼ / 1 + ¼ / 2 = 3/8
C	1/4	
D	1/4	

Cont.



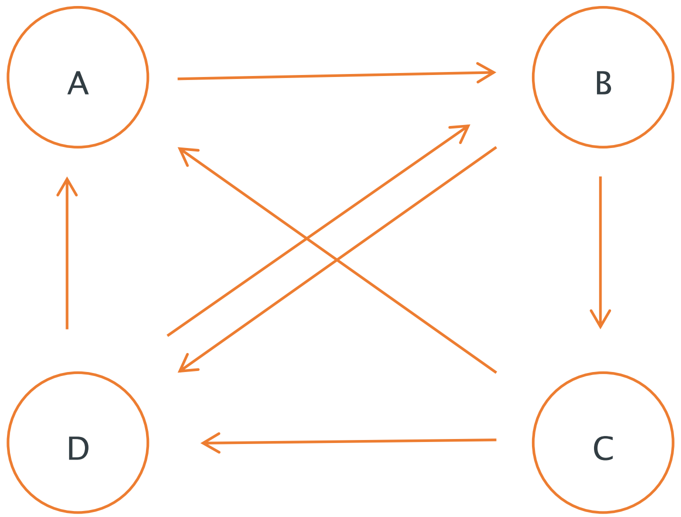
	Iteration 1	Iteration 2
A	1/4	¼ / 2 + ¼ / 2 = 2/8
B	1/4	¼ / 1 + ¼ / 2 = 3/8
C	1/4	¼ / 2 = 1/8
D	1/4	

Cont.



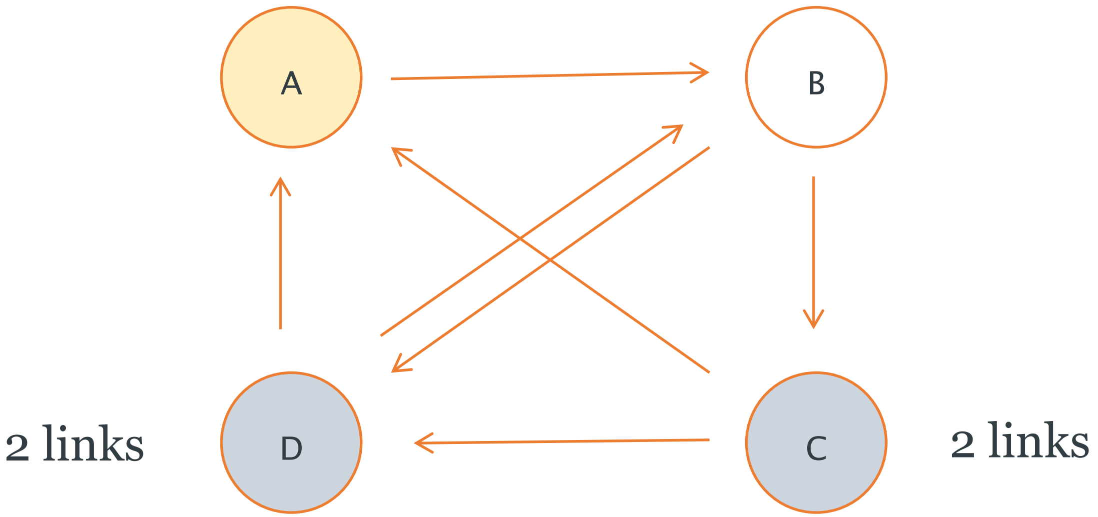
	Iteration 1	Iteration 2
A	1/4	¼ / 2 + ¼ / 2 = 2/8
B	1/4	¼ / 1 + ¼ / 2 = 3/8
C	1/4	¼ / 2 = 1/8
D	1/4	¼ / 2 + ¼ /2 = 2/8

Cont.



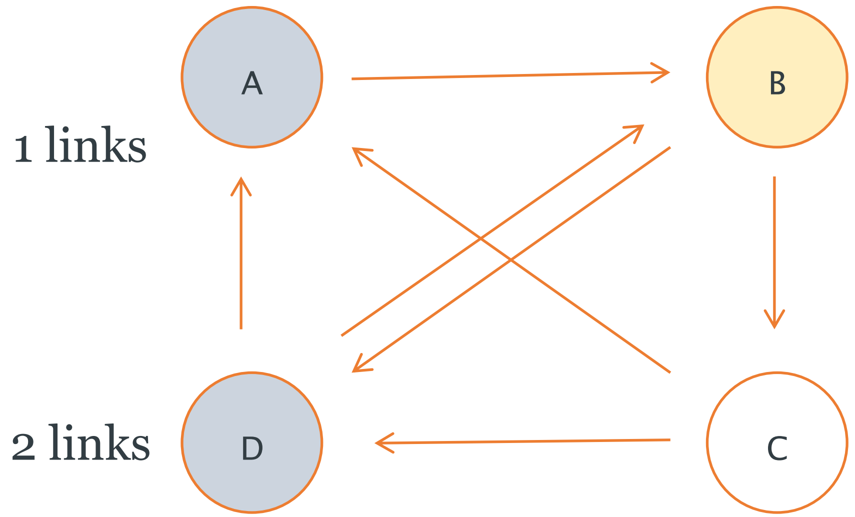
	Iteration 1	Iteration 2
A	1/4	2/8
B	1/4	3/8
C	1/4	1/8
D	1/4	2/8

Cont.



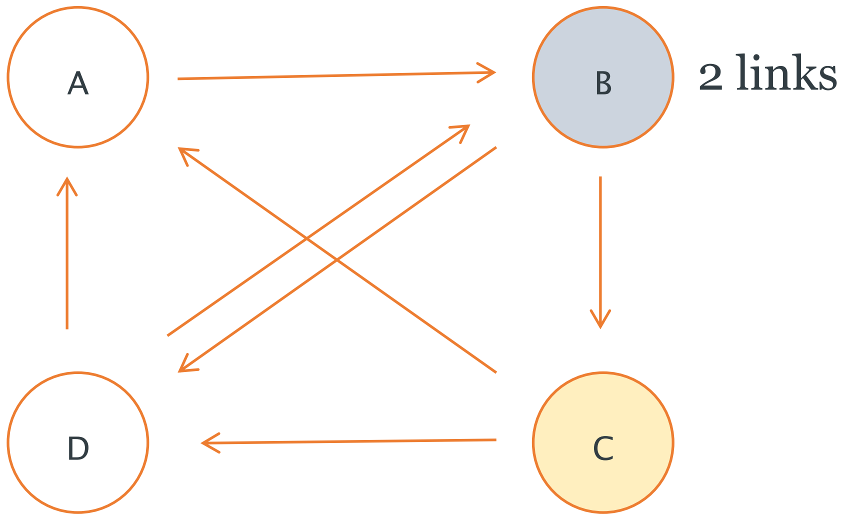
	Iteration 1	Iteration 2	Iteration 3
A	1/4	2/8	1/8 / 2 + 2/8 / 2 = 3/16
B	1/4	3/8	
C	1/4	1/8	
D	1/4	2/8	

Cont.



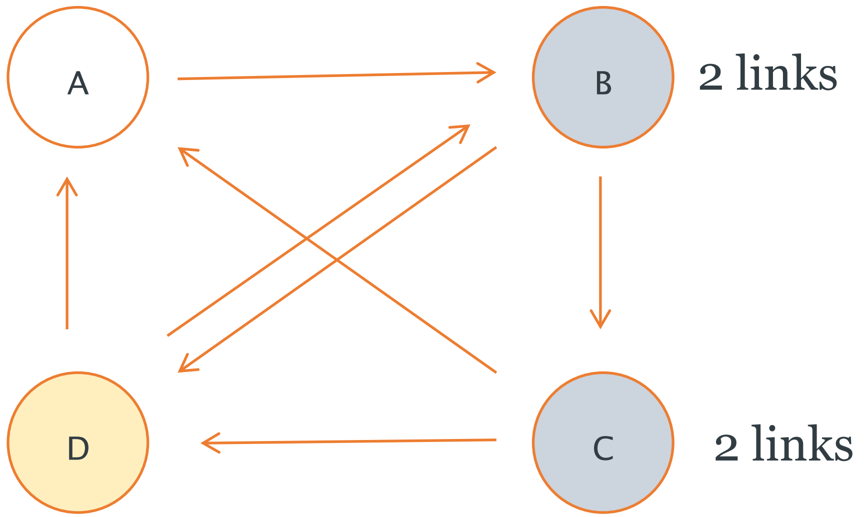
	Iteration 1	Iteration 2	Iteration 3		
A	1/4	2/8	1/8 / 2 + 2/8 / 2	=	3/16
B	1/4	3/8	2/8 / 1 + 2/8 / 2	=	6/16
C	1/4	1/8			
D	1/4	2/8			

Cont.



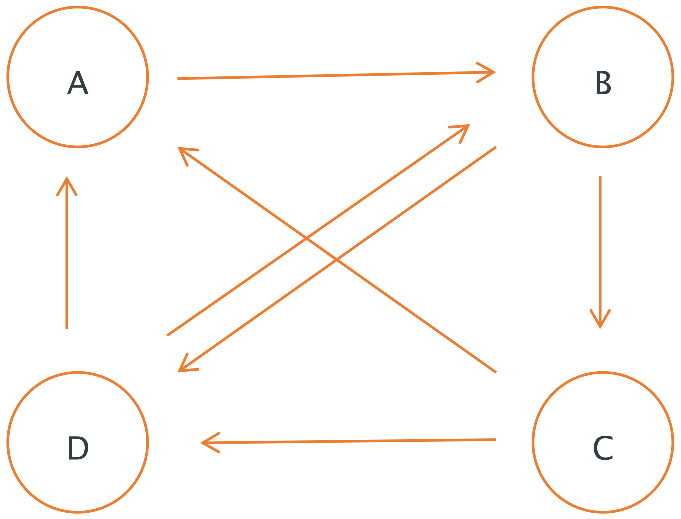
	Iteration 1	Iteration 2	Iteration 3		
A	1/4	2/8	1/8 / 2 + 2/8 / 2	=	3/16
B	1/4	3/8	2/8 / 1 + 2/8 / 2	=	6/16
C	1/4	1/8	3/8 / 2	=	3/16
D	1/4	2/8			

Cont.



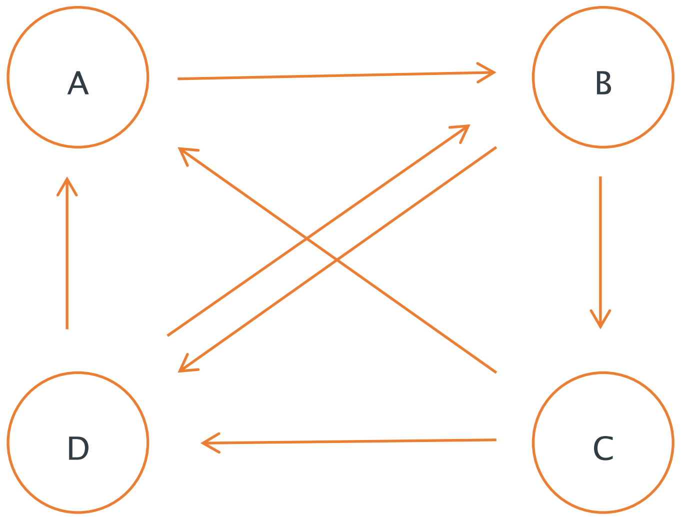
	Iteration 1	Iteration 2	Iteration 3
A	1/4	2/8	1/8 / 2 + 2/8 / 2 = 3/16
B	1/4	3/8	2/8 / 1 + 2/8 / 2 = 6/16
C	1/4	1/8	3/8 / 2 = 3/16
D	1/4	2/8	3/8 / 2 + 1/8 / 2 = 4/16

Cont.



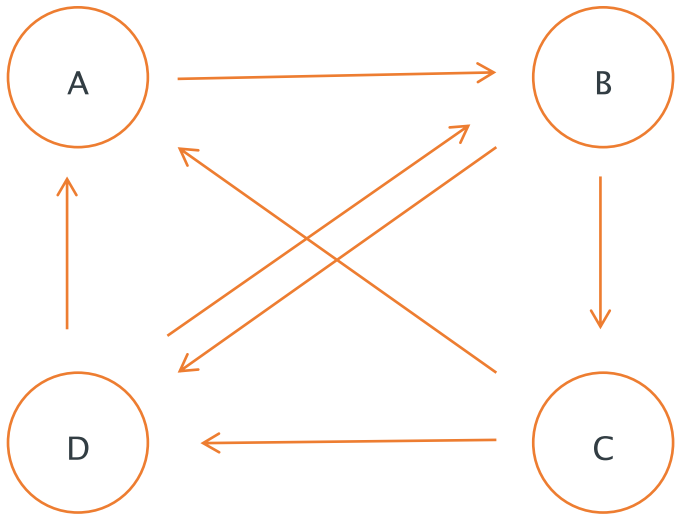
	Iteration 1	Iteration 2	Iteration 3
A	1/4	2/8	3/16
B	1/4	3/8	6/16
C	1/4	1/8	3/16
D	1/4	2/8	4/16

Cont.



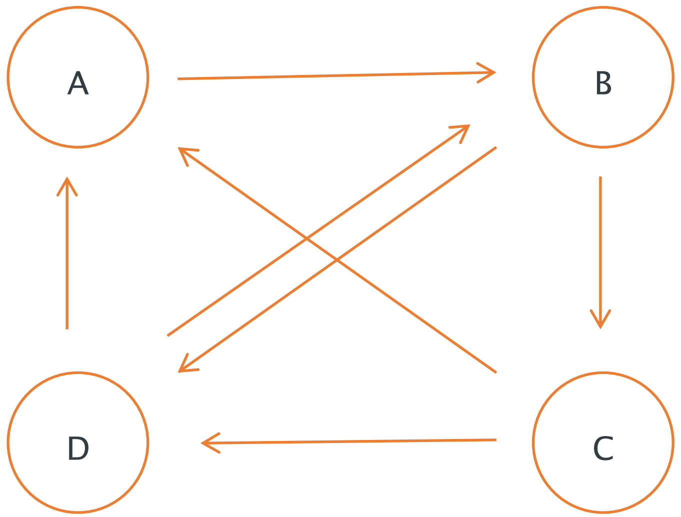
	Iteration 1	Iteration 2	Iteration 3	**Rank**
A	1/4	2/8	3/16	**3**
B	1/4	3/8	6/16	**1**
C	1/4	1/8	3/16	**3**
D	1/4	2/8	4/16	**2**

Cont.



	Iteration 3	Rank	**Iteration 10**	**Rank**
A	3/16	3	**0.2204**	**3**
B	6/16	1	**0.3505**	**1**
C	3/16	3	**0.1704**	**4**
D	4/16	2	**0.2585**	**2**

Convergence



	Iteration 1 Rank	Iteration 2 Rank	Iteration 3 Rank	Iteration 10 Rank
A	1	2	3	3
B	1	1	1	1
C	1	3	3	4
D	1	2	2	2

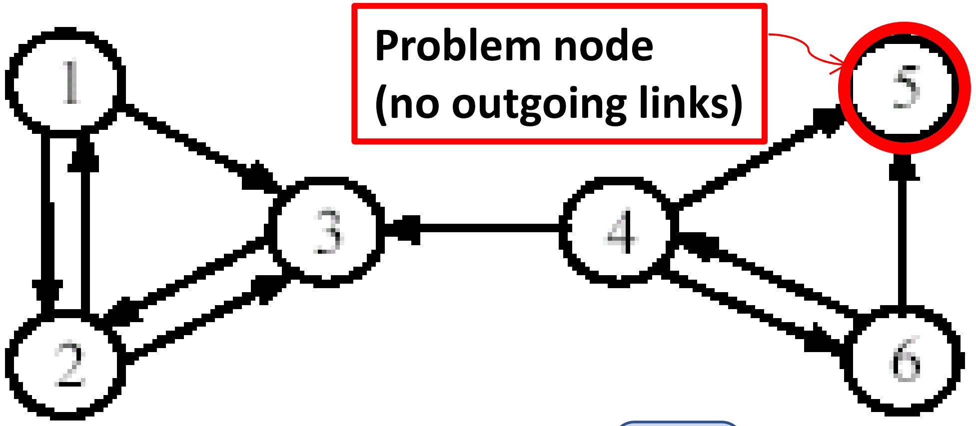
PageRank: some observations

- PR assumes important pages are likely be linked to them
- Similar to a voting:
  - where in(coming) links are votes
  - the quality of a vote is determined by the number of votes (in-links)
  - your vote is worth more if you have a higher page rank
- Link Quality
  - A link from an important page is worth more than many links from relatively unknown pages

Problem: Sink Nodes

- What happens when one of the nodes has indegree = 0 ?
  - Katz's "free" term β was added to all nodes
- Now, what if <u>outdegree = 0</u> ? → Sink
  - A sink is expressed as a column of zeros in the adjacency matrix

Sink problem



Recall that the problem with nodes with indegree = 0 was solved by using $\boldsymbol{\beta}$

in-degree matrix

$$\boldsymbol{A} = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$

each row shows the in degree

each column shows the out degree

$$x_i = \alpha \sum_j A_{ij} \frac{x_j}{out \deg j} + \beta$$

or in a *matrix from*

$$\boldsymbol{x}' = \alpha A D^{-1} \boldsymbol{x} + \beta \boldsymbol{1}$$

Inverse of the out-degree matrix

Fixing the sink problem

Add a complete set of outgoing links from each such page *i* to all other nodes/pages

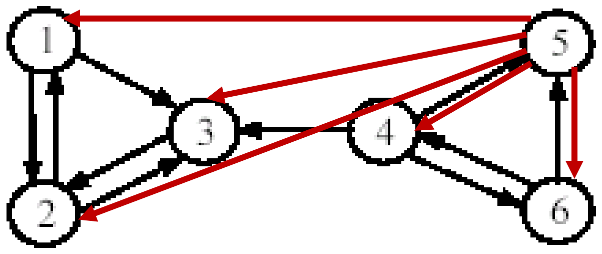
in-degree matrix

$$
\begin{array}{cc}
 & \text{each column is the out-degree} \\
A = & \begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}
\end{array}
$$

each row : In-degree

Fixing the out-degree = 0 sink problem

$$\boldsymbol{x}' = \alpha\, AD^{-1}\boldsymbol{x} + \beta\mathbf{1}$$



$$\boldsymbol{A} = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}$$

in-degree matrix

$$\boldsymbol{D}^{-1} = \begin{pmatrix} 1/2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1/2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1/1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/3 & 0 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1/6} & 0 \\ 0 & 0 & 0 & 0 & 0 & 1/2 \end{pmatrix}$$

Inverse of the out-degree matrix

Corrected PR centrality formula

By multiplying $\boldsymbol{A},\ \boldsymbol{D}^{-1}$ we obtain the matrix that captures:

$$\boldsymbol{x} = \alpha\, AD^{-1}\boldsymbol{x} + \beta\boldsymbol{1}$$

1. The in and out degree per vertex
2. Divides the centrality of each vertex by its degree

The contribution of node 5 is insignificant,

and the formula is now well defined

in-degree matrix

out-degree matrix

$$\boldsymbol{AD^{-1}} = \begin{pmatrix} 0 & 1/2 & 0 & 0 & \boxed{1/6} & 0 \\ 1/2 & 0 & 1 & 0 & \boxed{1/6} & 0 \\ 1/2 & 1/2 & 0 & 1/3 & \boxed{1/6} & 0 \\ 0 & 0 & 0 & 0 & \boxed{1/6} & 1/2 \\ 0 & 0 & 0 & 1/3 & \boxed{1/6} & 1/2 \\ 0 & 0 & 0 & 1/3 & \boxed{1/6} & 0 \end{pmatrix}$$

PR as a probabilistic model

- This modified AD^{-1} matrix is called the state transition probability matrix.
- Denote its entries by p_{ij} :

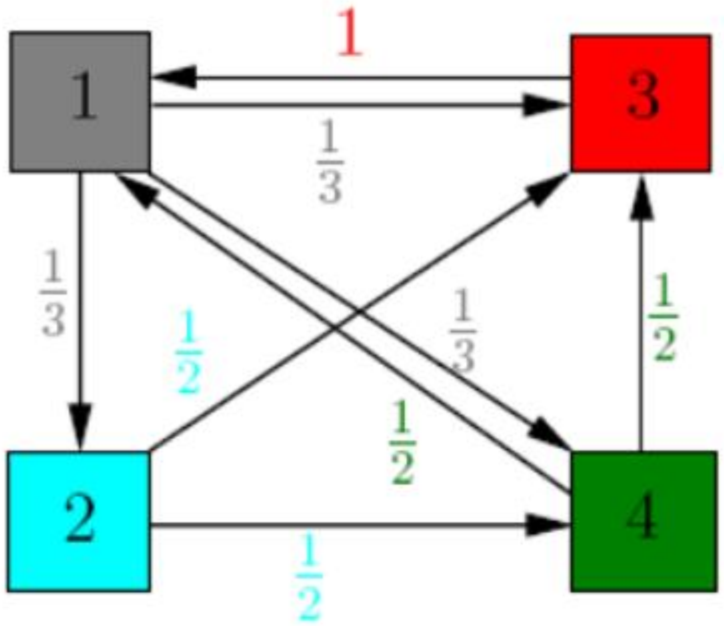
Transition probability matrix

$$AD^{-1} = \begin{pmatrix} p_{11} & p_{12} & \cdot & \cdot & \cdot & p_{1n} \\ p_{21} & p_{22} & \cdot & \cdot & \cdot & p_{2n} \\ \cdot & \cdot & & & & \cdot \\ \cdot & \cdot & & & & \cdot \\ \cdot & \cdot & & & & \cdot \\ p_{n1} & p_{n2} & \cdot & \cdot & \cdot & p_{nn} \end{pmatrix}$$

- p_{ij} represents the transition probability that the "surfer" in state j (page j) will move to state i (page i).

A small example of 4 websites (nodes)

p_{ij} represents the transition probability that the surfer on page j will move to page i:



$$AD^{-1} = (p_{ij}) = \begin{pmatrix} 0 & 0 & 1 & 1/2 \\ 1/3 & 0 & 0 & 0 \\ 1/3 & 1/2 & 0 & 1/2 \\ 1/3 & 1/2 & 0 & 0 \end{pmatrix}$$

Small example cont.

<u>Random surfer</u>:

Each page has equal probability ¼ to be chosen as a starting point.

$$v = \begin{pmatrix} 0.25 \\ 0.25 \\ 0.25 \\ 0.25 \end{pmatrix}, \quad Av = \begin{pmatrix} 0.37 \\ 0.08 \\ 0.33 \\ 0.20 \end{pmatrix}, \quad A^2 v = A(Av) = A \begin{pmatrix} 0.37 \\ 0.08 \\ 0.33 \\ 0.20 \end{pmatrix} = \begin{pmatrix} 0.43 \\ 0.12 \\ 0.27 \\ 0.16 \end{pmatrix}$$

$$A^3 v = \begin{pmatrix} 0.35 \\ 0.14 \\ 0.29 \\ 0.20 \end{pmatrix}, \quad A^4 v = \begin{pmatrix} 0.39 \\ 0.11 \\ 0.29 \\ 0.19 \end{pmatrix}, \quad A^5 v = \begin{pmatrix} 0.39 \\ 0.13 \\ 0.28 \\ 0.19 \end{pmatrix}$$

$$A^6 v = \begin{pmatrix} 0.38 \\ 0.13 \\ 0.29 \\ 0.19 \end{pmatrix}, \quad A^7 v = \begin{pmatrix} 0.38 \\ 0.12 \\ 0.29 \\ 0.19 \end{pmatrix}, \quad A^8 v = \begin{pmatrix} 0.38 \\ 0.12 \\ 0.29 \\ 0.19 \end{pmatrix}$$

**** A here is AD^{-1}**

Small example cont.

- The probability that page *i* will be visited after *k* steps (i.e. the random surfer ending up at page *i*) is equal to i^{th} entry of $A^k x$.
- The sequence of iterations $v, Av, \ldots, A^k v$ converges to the equilibrium value

$$v^* = \begin{pmatrix} 0.38 \\ 0.12 \\ 0.29 \\ 0.19 \end{pmatrix}$$

- This is the PageRank vector of this (web) graph
- Note that this simplified example had no β since all in-deg >0

PageRank Algorithm

PageRank of Node X

$$PR(x) = \frac{(1-d)}{N} + d\sum_{y->x}\frac{PR(y)}{L(y)}$$

Assuming there are NO Sink nodes

- Page Rank of Node X is the probability of being at node X at the current time.
- How can we visit node X from where we are?
  - (1-d) term: Random Jump: The probability of ending up at node X because of a random jump from some node, including node X, is 1/N.
  - However, such a random jump itself could occur with a probability of (1-d).
  - (1-d)/N is the probability to be at node X due to a random jump.

PageRank Algorithm

d term: Edge Traversal from a Neighbor:

- We could visit node X from one of the nodes that point to node X.
- Lets say, we are at node Y in the previous iteration. The probability of being at node Y in the previous iteration is PR(Y). We can visit any of Y's neighbors.
- The probability of visiting node X among the L(y) out-going links of Y is PR(Y) / L(Y).
- Likewise, we could visit X from any of its neighbors.
- All the probabilities of visiting X from any of its neighbors have to be added.
- The whole event of visiting from a neighbor occurs with a prob. 'd'

PageRank Algorithm (with) Sink Nodes

- **Assume that the sink node is connected to all other nodes (incl. self).**
  - **The sink node equally shares its PR with every node in the graph, including itself.**
  - If *z* is a sink node, with the above scheme, out(z) = N, the number of nodes in the graph.
- **The prob. of getting to node X at a given time is the two terms below:**
  - Random jump from any node (probability, 1-d)
  - Visit from a node with in-link to node X (probability, d)

Page Rank of Node X

$$PR(x) = \frac{(1-d)}{N} + d\sum_{y \to x}\frac{PR(y)}{L(y)} + d\sum_{z \to \varphi}\frac{PR(z)}{N}$$

Explicit out-going links to certain nodes

Implicit out-going links to all nodes (sink nodes)

Hypertext Induced Topic Search (HITS)

- **HITS Centrality**

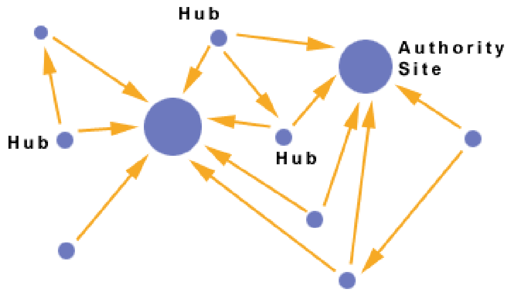
HITS Centrality – Hubs and Authorities

- Belongs to Spectral measures
- HITS Centrality: A node is more important if it has more links
- Hubs and Authorities

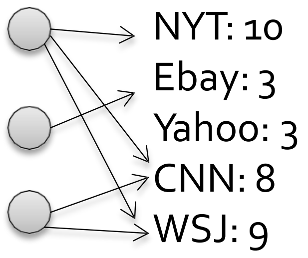
- Each node has 2 scores:
  - Quality as an expert (hub):
    - Total sum of votes of node that it points to
  - Quality as a content provider (authority):
    - Total sum of votes from hubs/experts

Interesting pages fall into two classes: hubs & auth.

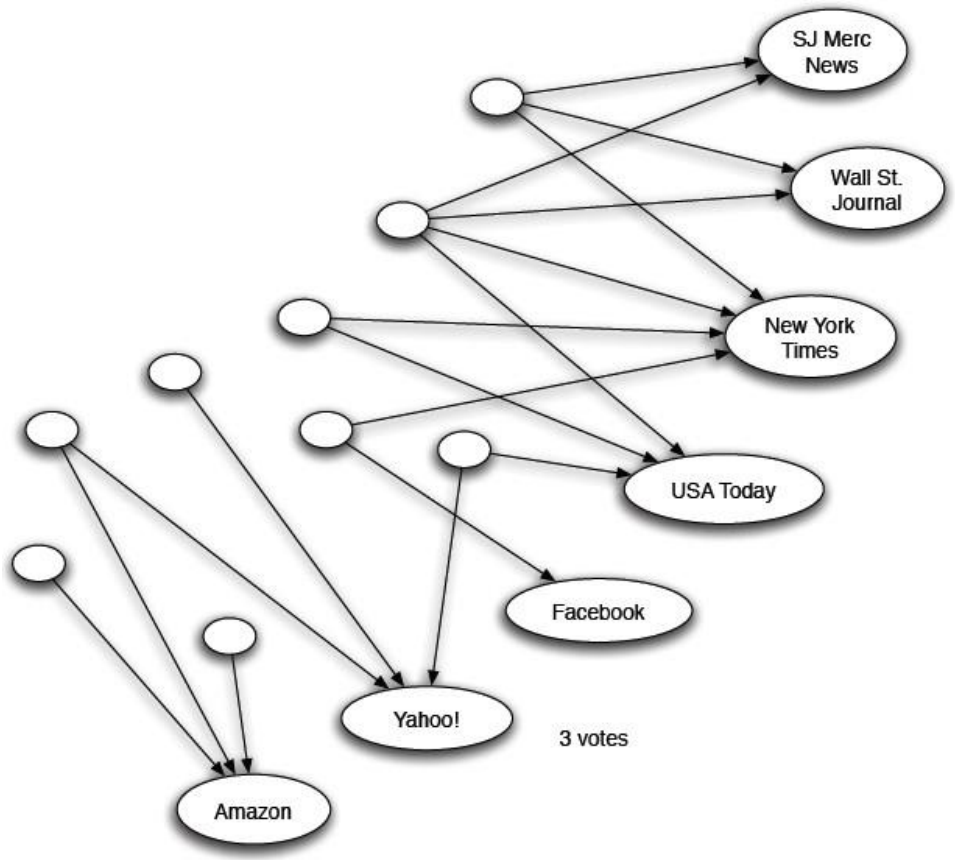
- Authorities are nodes containing useful information
  - Newspaper home pages
  - Course home pages
  - Home pages of auto manufacturers



- Hubs are nodes that link to authorities
  - List of newspapers
  - Course newspaper
  - List of auto manufacturers



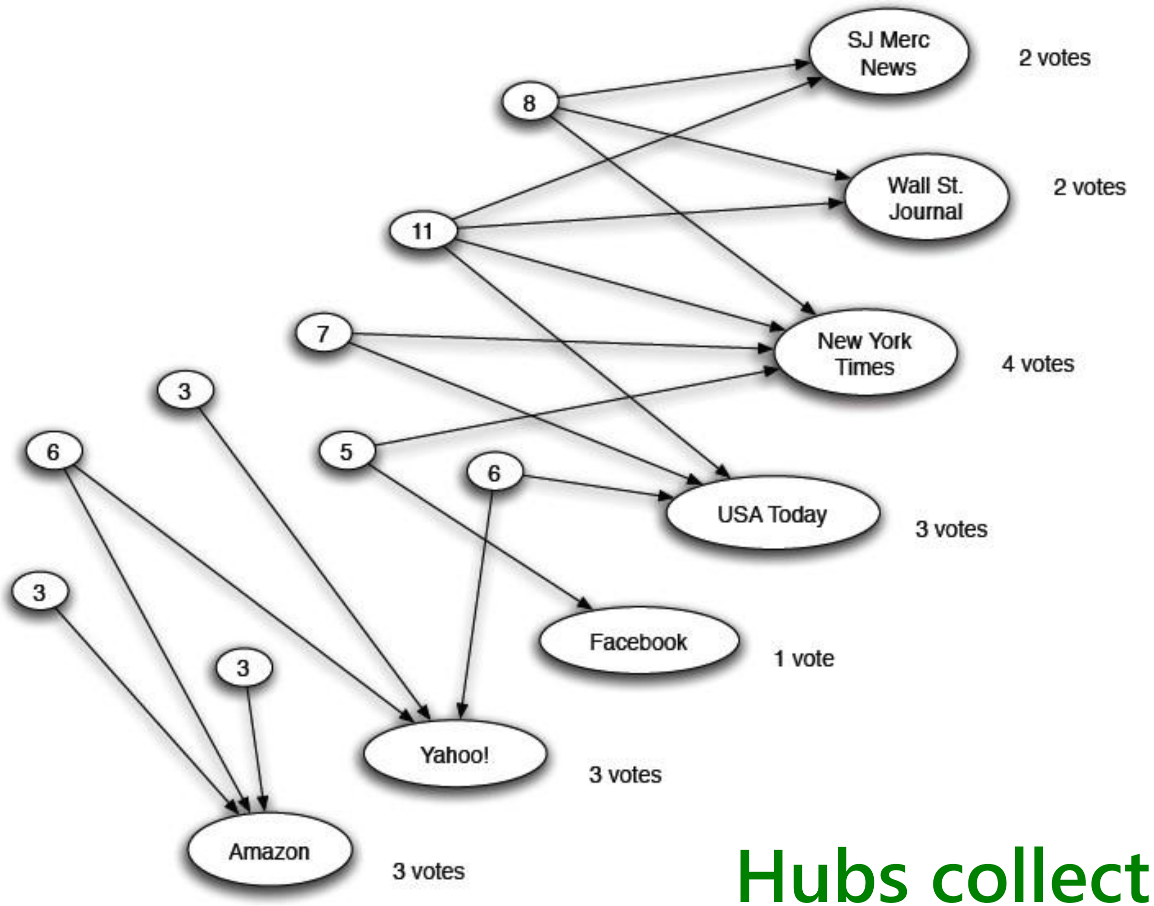
Counting in links = Authority



Each page starts with hub score 1
Authorities collect their votes

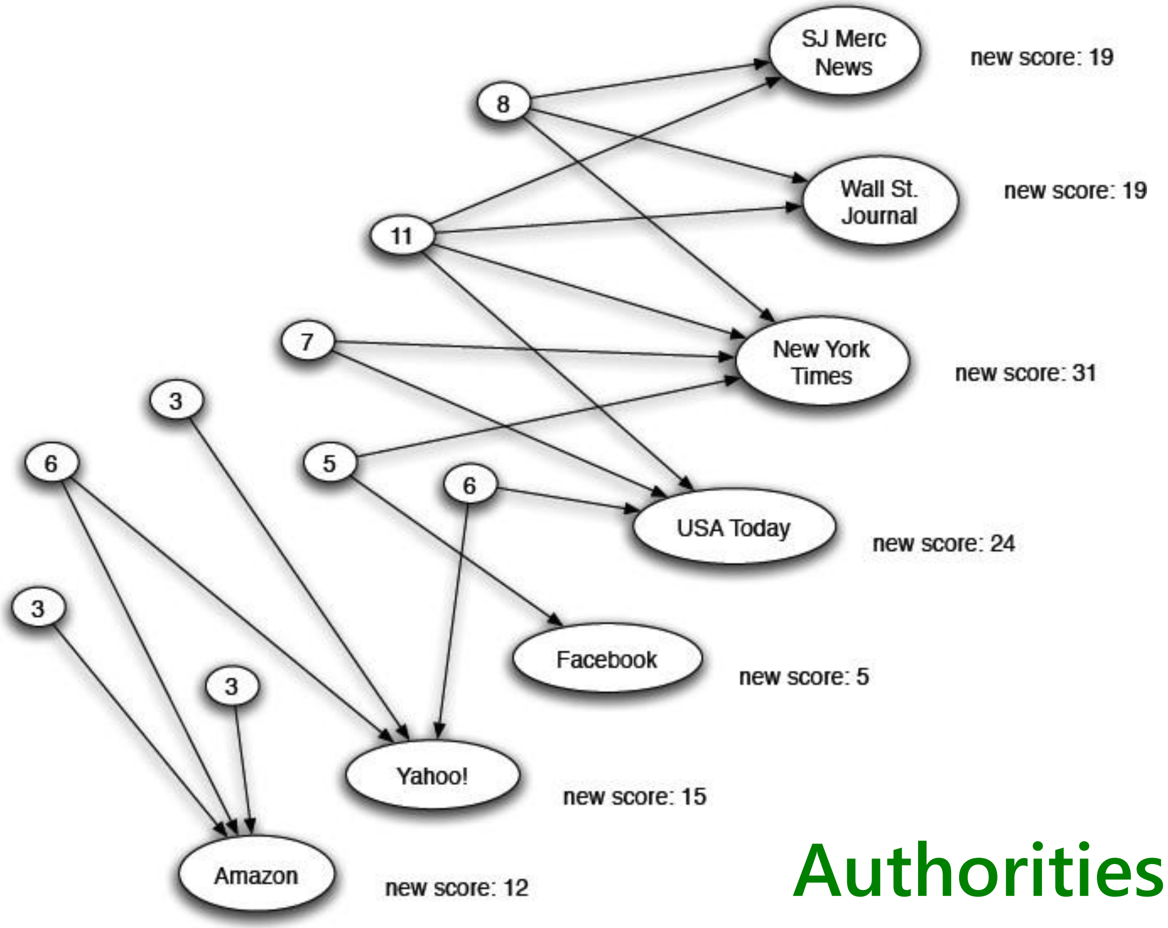
Note this is an idealized example. In reality graph is not bipartite and each page has both the hub and the authority score.

Expert Quality = Hub



Hubs collect authority scores

Reweighting



Authorities collect hub scores

Authorities and Hubs

- *Authorities* are pages (nodes) that are recognized as providing significant, trustworthy, and useful information on a topic.
  - *In-degree* (number of pointers to a page) is one simple measure of authority.
  - However in-degree treats all links as equal.
  - Should links from pages that are themselves authoritative count more?
- *Hubs* are index pages that provide lots of useful links to relevant content pages (topic authorities).

HITS

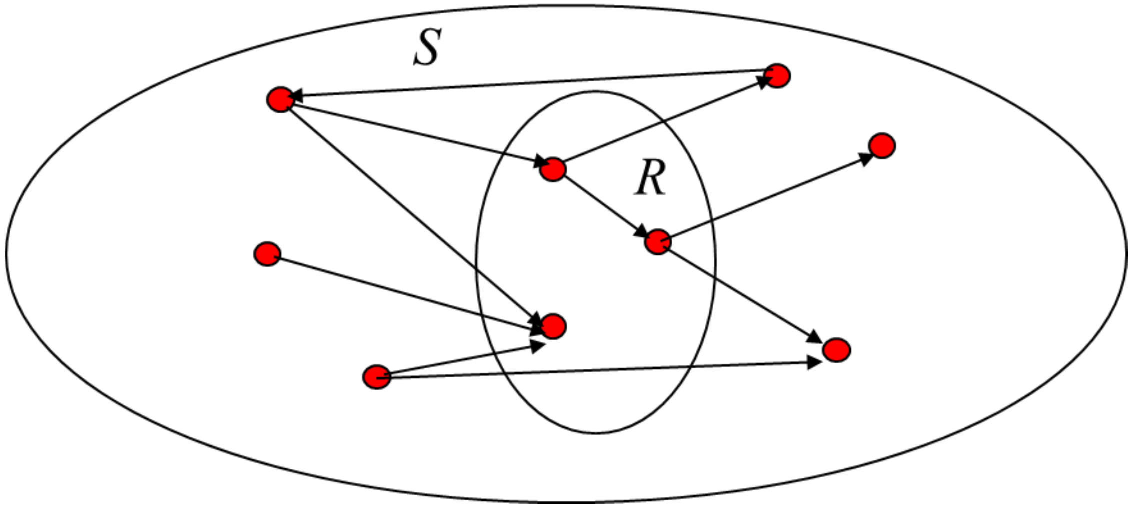
- Algorithm developed by Kleinberg in 1998 (around the same time as PageRank)
- Attempts to computationally determine hubs and authorities on a particular topic through analysis of a relevant subgraph of the web.
- Based on mutually recursive facts:
  - Hubs point to lots of authorities.
  - Authorities are pointed to by lots of hubs.

HITS Algorithm

- Computes hubs and authorities for a particular topic specified by a query.
- First determines a set of relevant pages for the query called the *base* set *S*.
- Analyze the link structure of the web subgraph defined by *S* to find authority and hub pages in this set.

Constructing a Base Subgraph

- For a specific query *Q*, let the set of documents returned by a standard search engine be called the *root* set *R*.
- Initialize *S* to *R*.
- Add to *S* all pages pointed to by any page in *R*.
- Add to *S* all pages that point to any page in *R*.



Iterative Algorithm

- Use an iterative algorithm to slowly converge on a mutually reinforcing set of hubs and authorities.
- Maintain for each page/node $p \in S$:
  - Authority score: a_p (vector a)
  - Hub score: h_p (vector h)
- Initialize all $a_p = h_p = 1$
- Maintain normalized scores:

$$\sum_{p \in S} (a_p)^2 = 1 \qquad \sum_{p \in S} (h_p)^2 = 1$$

HITS Update Rules

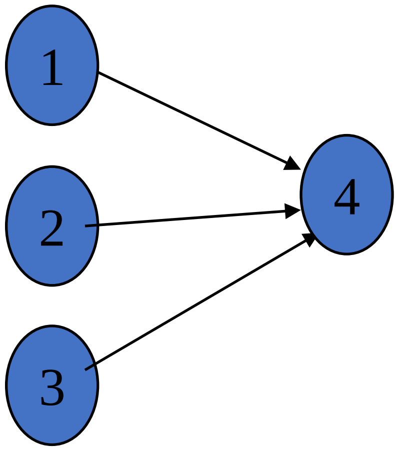
- Authorities are pointed to by lots of good hubs:

$$a_p = \sum_{q:q \to p} h_q$$

- Hubs point to lots of good authorities:

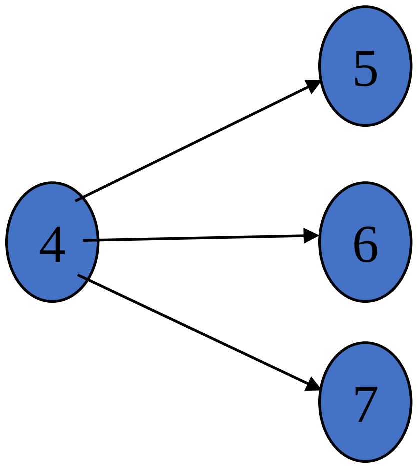
$$h_p = \sum_{q:p \to q} a_q$$

Illustrated Update Rules



$a_4 = h_1 + h_2 + h_3$

$h_4 = a_5 + a_6 + a_7$



HITS Iterative Algorithm

Initialize for all $p \in S$: $a_p = h_p = 1$

For $i = 1$ *to* k:

For all $p \in S$: $$a_p = \sum_{q:q \to p} h_q$$ *(update auth. scores)*

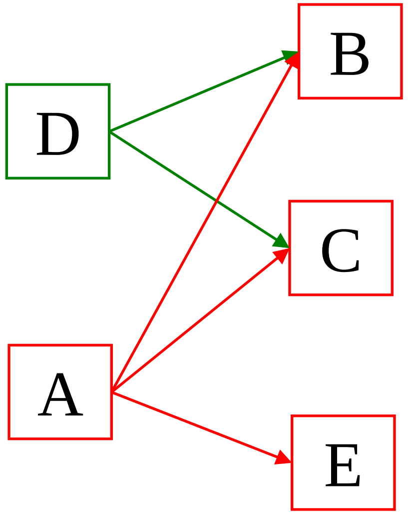
For all $p \in S$: $$h_p = \sum_{q:p \to q} a_q$$ *(update hub scores)*

For all $p \in S$: $a_p = a_p / c$ c: $$\sum_{p \in S} (a_p / c)^2 = 1$$ *(normalize a)*

For all $p \in S$: $h_p = h_p / c$ c: $$\sum_{p \in S} (h_p / c)^2 = 1$$ *(normalize h)*

C = square root of the sum of the squares

HITS Example



First Iteration

	D	A	C	B	E
A:	[0.0,	0.0,	2.0,	2.0,	1.0]

	D	A	C	B	E
H:	[4.0,	5.0,	0.0,	0.0,	0.0]

Normalize: divide each vector by its norm (square root of the sum of the squares)

	D	A	C	B	E
Normalized A:	[0.0,	0.0,	0.67,	0.67.0,	0.33]

	D	A	C	B	E
Norm. H:	[0.62,	0.78,	0.0,	0.0,	0.0]

Norm(A)= sqrt(4+4+1)=3

Convergence

- Algorithm converges to a *fix-point* if iterated indefinitely.
- Define A to be the adjacency matrix for the <u>subgraph defined by *S*</u>.
  - $A_{ij} = 1$ for $i \in S$, $j \in S$ iff $i \rightarrow j$
- Authority vector, a, converges to the principal eigenvector of $A^T A$
- Hub vector, h, converges to the principal eigenvector of AA^T
- In practice, 20 iterations produces fairly stable results.

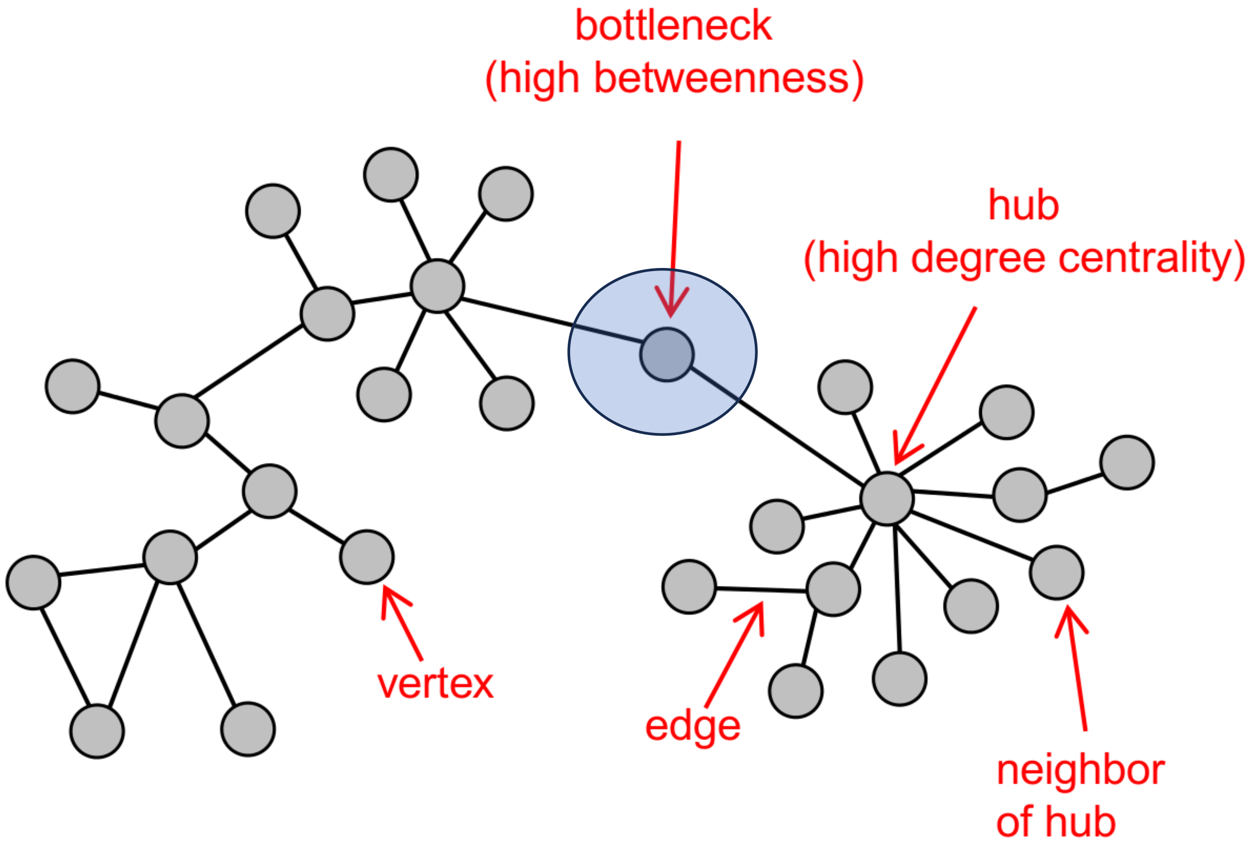
Path-Based Measures

- **Betweenness**
  - Node
  - Vertex

Intuition

- How many pairs of individuals would have to go through you in order to reach one another in the minimum number of hops?
- Interactions between two individuals depend on the other individuals in the set of nodes. The nodes in the middle have some control over the paths in the graph.
- Useful for flow, such as information or data packages

Intuition

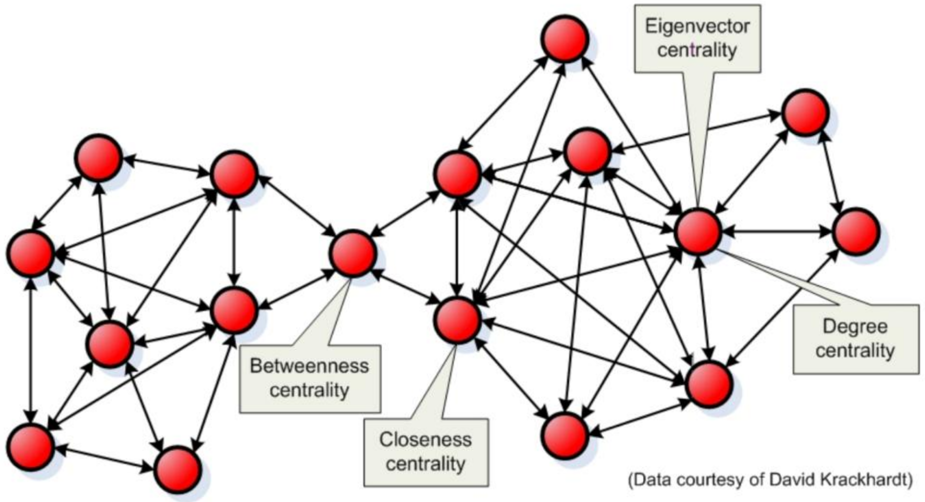


bottleneck
(high betweenness)
hub
(high degree centrality)
vertex
edge
neighbor
of hub

Meaning of betweenness centrality

Vertices with high betweenness centrality have influence in the network because they control information passing between others.

- **They get to "see" the messages as they pass through**
- **Thus their removal would disrupt communication**



Betweenness centrality

$$g(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$$

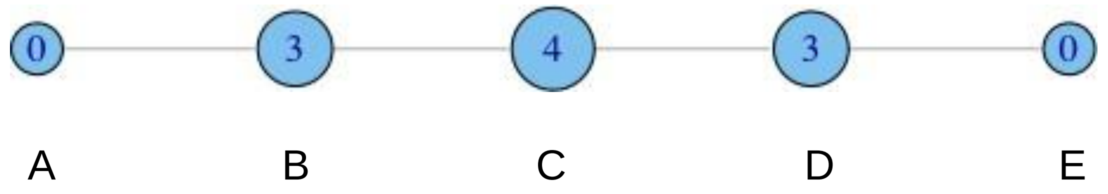
- σ_{st}^{v} is the number of *s-t* geodesics that pass through *v*

  (default: *v* could equal *s* or *t*, but in other versions it cannot and that's where you see 0 values)

- σ_{st} is the total number of geodesics between s-t
- In an undirected graph, an *s-t* geodesic is the same as a *t-s* geodesics, so the edge gets counted twice
- It is applicable to directed networks as well.

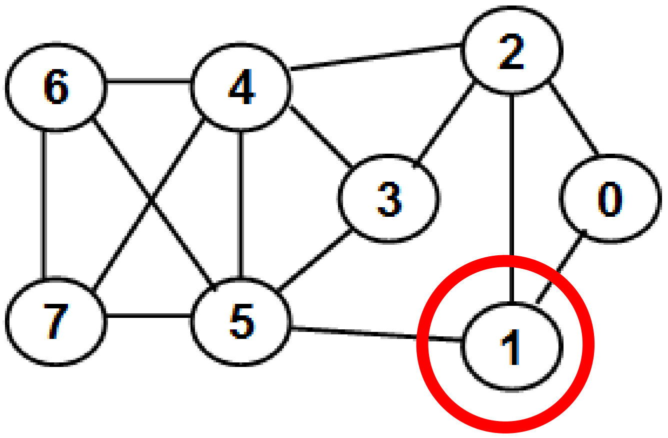
Betweenness Centrality Example

- Removing nodes in betweenness order causes a very quick disruption of the network



- A lies between no two other vertices
- B lies between A and 3 other vertices: C, D, and E
- C lies between 4 pairs of vertices (A,D),(A,E),(B,D),(B,E)
- note that there are no alternate paths for these pairs to take, so C gets full credit

Betweenness Centrality Example 2



<u>BWC for node 0</u>: 0.0

<u>BWC for node 1</u>
Pair (0, 5): → 1 / 1
Pair (0, 6): → 1 / 2
Pair (0, 7): → 1 / 2
Pair (2, 5): → 1 / 3
BWC (1) = 2.333

<u>BWC for node 2</u>
Pair (0, 3): → 1 / 1
Pair (0, 4): → 1 / 1
Pair (0, 6): → 1 / 2
Pair (0, 7): → 1 / 2
Pair (1, 3): → 1 / 2
Pair (1, 4): → 1 / 2
BWC (2): 4.0

<u>BWC for node 3</u>
Pair (2, 5) → 1 / 3
BWC (3) = 0.333

<u>BWC for node 4</u>
Pair (0, 6) → 1 / 2
Pair (0, 7) → 1 / 2
Pair (2, 5) → 1 / 3
Pair (2, 6) → 1 / 1
Pair (2, 7) → 1 / 1
Pair (3, 6) → 1 / 2
Pair (3, 7) → 1 / 2
BWC (2) = 4.333

<u>BWC for node 6:</u> 0.0
<u>BWC for node 7:</u> 0.0

<u>BWC for node 5</u>
Pair (0, 6) → 1 / 2
Pair (0, 7) → 1 / 2
Pair (1, 3) → 1 / 2
Pair (1, 4) → 1 / 2
Pair (1, 6) → 1 / 1
Pair (1, 7) → 1 / 1
Pair (3, 6) → 1 / 2
Pair (3, 7) → 1 / 2
BWC (5) = 5.0

ID	BWC
0	0.0
1	2.333
2	4.0
3	0.333
4	4.333
5	5.0
6	0.0
7	0.0

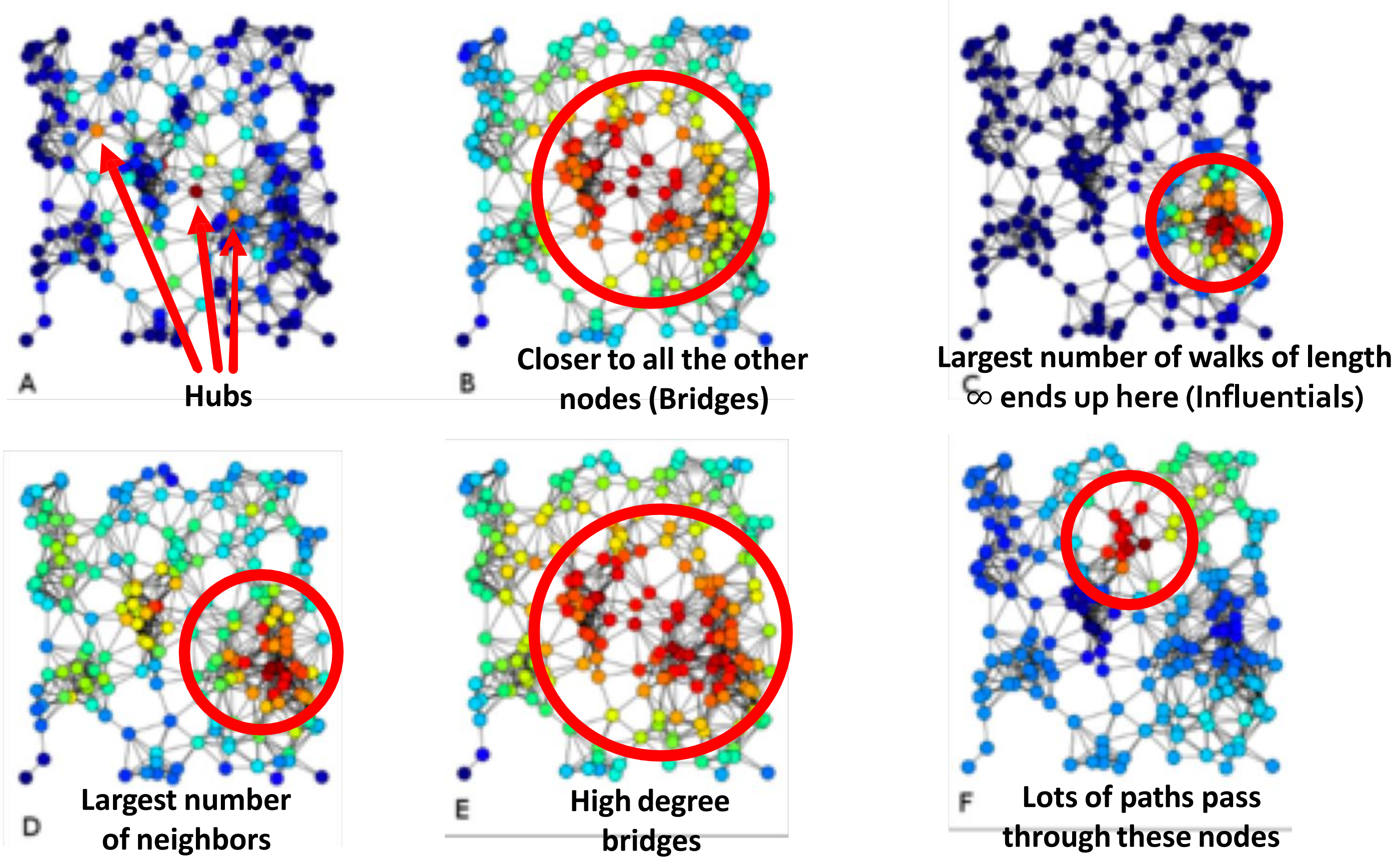
A normalized refined formula

- **Normalized relative betweenness centrality allows to compare nodes in other graphs.**

  - $x_i = \left(\sum \frac{n_{st}^i}{g_{st}}\right) / n^2$

- **where**
  - n_{st}^i **is the number of** ***s-t*** **geodesics that** ***i*** **belongs to.**
  - g_{st} **is the number of** ***s-t*** **geodesics**
  - **Convention:** g_{st} **= 0 and** n_{st}^i **= 0, then** $x_i = 0$

Network Centrality Example



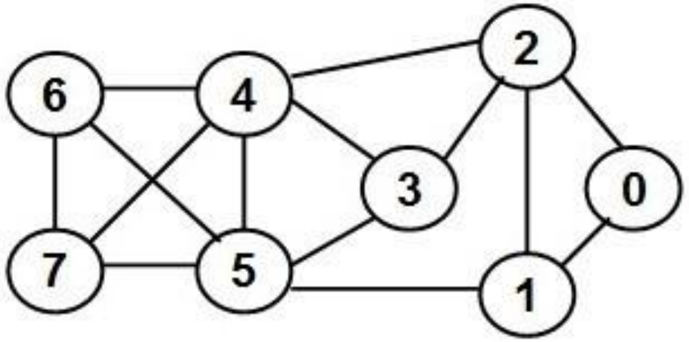
A) Betweenness centrality, B) Closeness centrality, C) Eigenvector centrality, D) Degree centrality, E) Harmonic Centrality and F) Katz centrality of the same graph.

EGO Networks

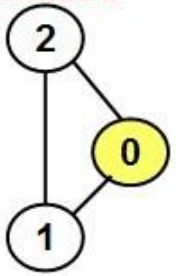
Ego Network

- The ego network is node-specific.
- The ego network for a node comprises of the node and its neighbors as vertices and the links connecting the node and/or its neighbors as edges.
- The BWC of a vertex computed on the entire graph is directly related to the BWC of the vertex computed on its ego network.
- Could be used to rank the vertices.

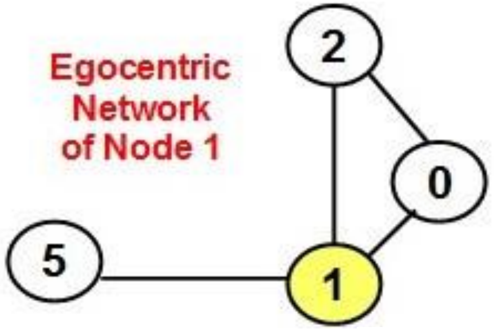
Ego Network

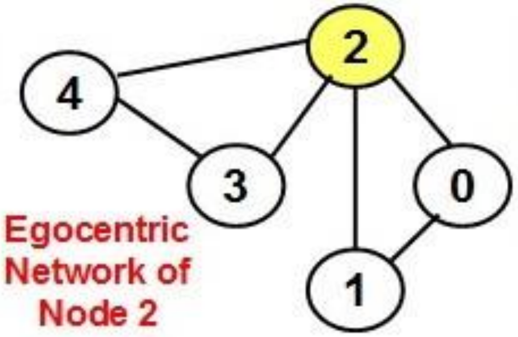


Egocentric Network
of Node 0



Egocentric
Network
of Node 1





Egocentric
Network of
Node 2

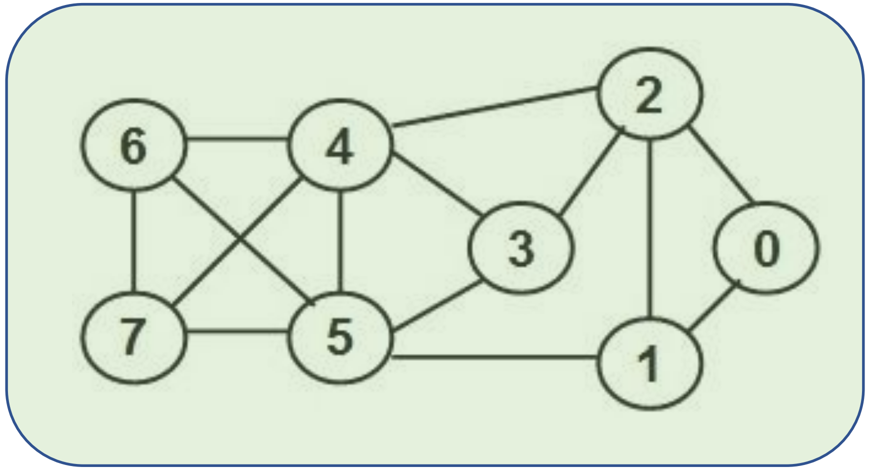
Ego BWC

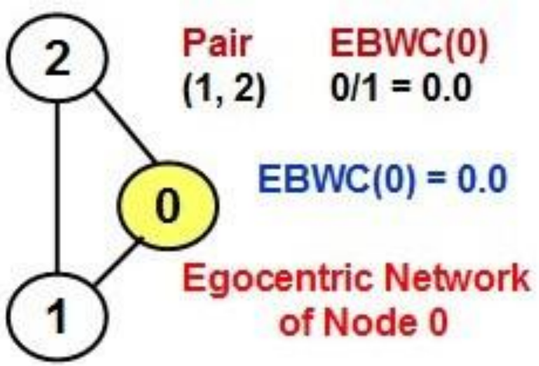
- The Ego BWC of a vertex is the BWC of the vertex computed on its ego centric network graph.
- A vertex that is not on the shortest path connecting any of its neighbor nodes is definitely not needed to be on the shortest path for any other pair of nodes in the network.
- On the other hand, a vertex that is on the shortest path for several pairs of its neighbor nodes is more likely to be also on the shortest path for several other node pairs (at least in the vicinity of the node).

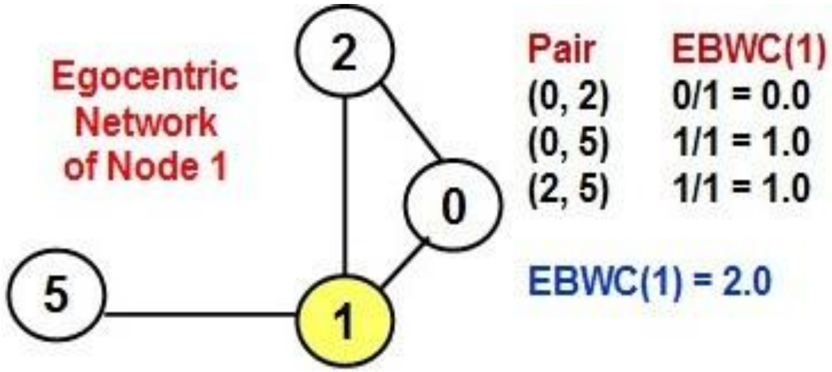
Ego BWC example

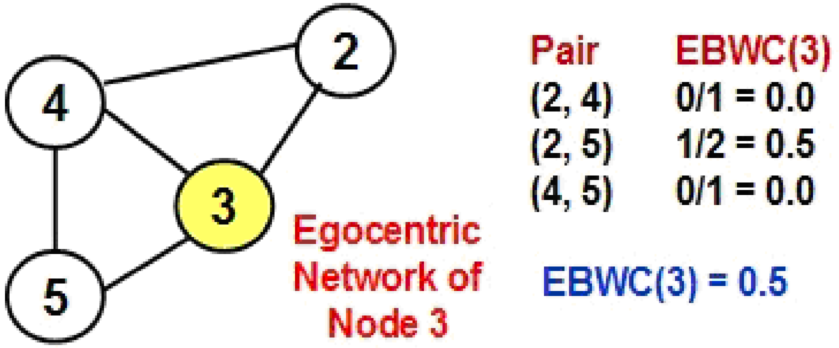
Calculate EBWC

for each node

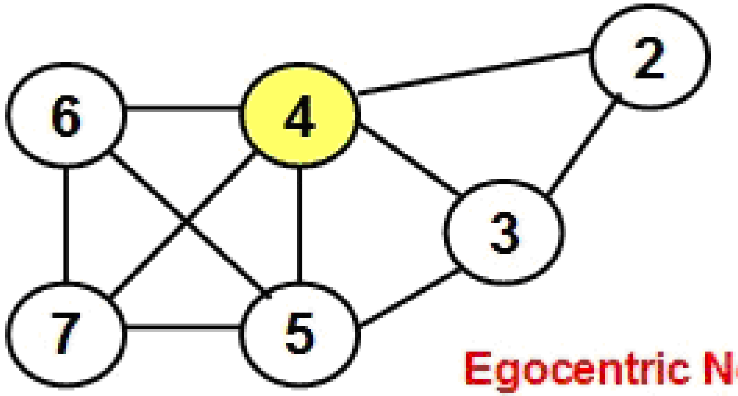








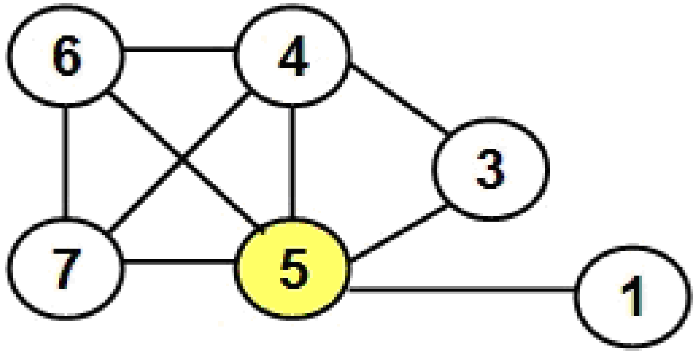
Ego BWC



Egocentric Network of Node 4

Pair	EBWC(4)	Pair	EBWC(4)
(2, 3)	0/1 = 0.0	(3, 6)	1/2 = 0.5
(2, 5)	1/2 = 0.5	(3, 7)	1/2 = 0.5
(2, 6)	1/1 = 1.0	(5, 6)	0/1 = 0.0
(2, 7)	1/1 = 1.0	(5, 7)	0/1 = 0.0
(3, 5)	0/1 = 0.0	(6, 7)	0/1 = 0.0

EBWC(4) = 3.5

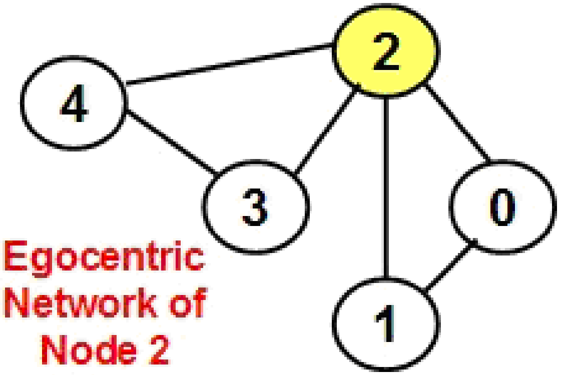


Egocentric Network of Node 5

Pair	EBWC(5)	Pair	EBWC(5)
(1, 3)	1/1 = 1.0	(3, 6)	1/2 = 0.5
(1, 4)	1/1 = 1.0	(3, 7)	1/2 = 0.5
(1, 6)	1/1 = 1.0	(4, 6)	0/1 = 0.0
(1, 7)	1/1 = 1.0	(4, 7)	0/1 = 0.0
(3, 4)	0/1 = 0.0	(6, 7)	0/1 = 0.0

EBWC(5) = 5.0

Ego BWC

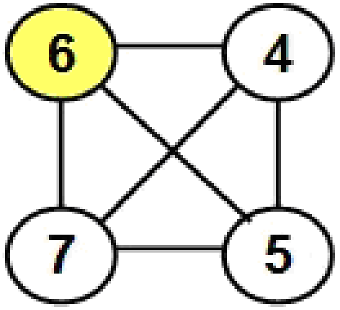


Egocentric Network of Node 2

Pair	EBWC(2)
(0, 1)	0/1 = 0.0
(0, 3)	1/1 = 1.0
(0, 4)	1/1 = 1.0
(1, 3)	1/1 = 1.0
(1, 4)	1/1 = 1.0
(3, 4)	0/1 = 0.0

EBWC(2) = 4.0

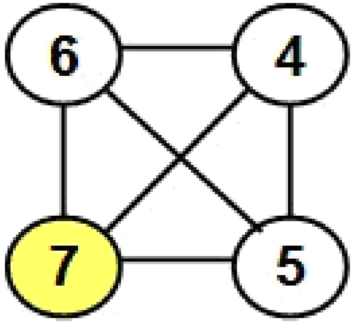
ID	EBWC	BWC
0	0.0	0.0
1	2.0	2.33
2	4.0	4.0
3	0.5	0.33
4	3.5	4.33
5	5.0	5.0
6	0.0	0.0
7	0.0	0.0



Pair	EBWC(6)
(4, 5)	0/1 = 0.0
(4, 7)	0/1 = 0.0
(5, 7)	0/1 = 0.0

EBWC(6) = 0.0

Egocentric Network of Node 6



Pair	EBWC(7)
(4, 5)	0/1 = 0.0
(4, 6)	0/1 = 0.0
(5, 6)	0/1 = 0.0

EBWC(7) = 0.0

Egocentric Network of Node 7